VITCC CENTRAL LIBRARY MANAGEMENT SYSTEM

MULTI-ROLE INTERFACE AUDIT & ROLE-BASED ACCESS CONTROL (RBAC) VERIFICATION REPORT

Automated high-fidelity capture of all system sections across Administrator, Librarian, Faculty/Staff, Student, and Public portals for autonomous LLM audit and compliance review.

Audit Date: October 18, 2026	**Target Environment:** Live Dev Server (http://127.0.0.1:5000)	**Display Viewport:** 1440 x 960 (Desktop HD)
Database Engines: Oracle Database 23ai + Neo4j Graph DB	**Total Views Audited:** 44 Unique Verified Screens	**Security Model:** Session RBAC + CSRF Protection

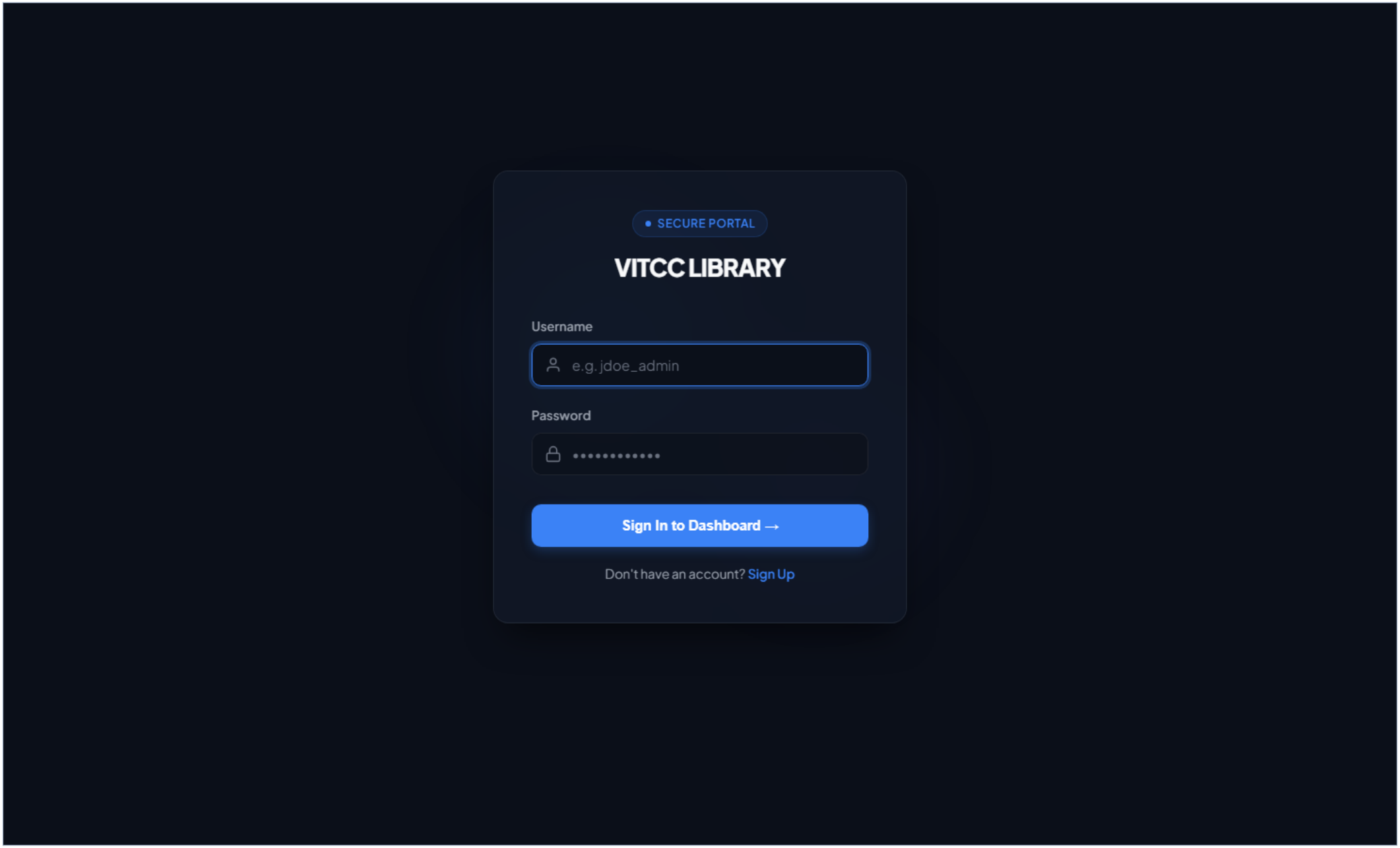
Application Section / Feature	Route URL	Public	Admin	Librarian	Staff	Student	Access Scope / Security Rule
Login & Authentication	/login	✓	✓	✓	✓	✓	Public credential verification & session issuing
Patron Registration	/register	✓	✓	✓	✓	✓	Self-service patron sign up (Student/Staff)
Dashboard & Overview	/dashboard	—	**✓ Full**	**✓ Desk**	**✓ Faculty**	**✓ Personal**	Role-tailored KPI cards, borrowing queues & charts
Catalog & Book Search	/books	—	✓	✓	✓	✓	Global catalog search with category & author filtering
Category & Curriculum Graph	/category/<id>	—	✓	✓	✓	✓	Prerequisite & unlocks badges, co-borrowing pills
Issue Book Loan	/issue_loan	—	**✓ Staffed**	**✓ Staffed**	**Blocked**	**Blocked**	Direct copy allocation desk (Admin & Librarian only)
Loan Requests & Collection	/loan_requests	—	**✓ Staffed**	**✓ Staffed**	**Blocked**	**Blocked**	Patron pickup approvals & collection desk handover
Manage Users & Governance	/manage_users	—	**✓ Admin**	**Blocked**	**Blocked**	**Blocked**	RBAC privileges, suspension & password resets (Admin)
Learning Roadmaps	/learning_paths	—	✓	✓	✓	✓	Curriculum tracks & multi-level study hierarchies
AI Recommendations	/recommendations	—	**✓ Trends**	**✓ Trends**	**✓ Faculty**	**✓ Peer**	Collaborative filtering, peer study partners & trends
My Fines & Settlements	/my_fines	—	—	—	**✓ Faculty**	**✓ Student**	Personal overdue fines & instant payment gateway
Knowledge Graph Explorer	/graph_visualizer	—	**✓ Global**	**✓ System**	**✓ Academic**	**✓ Learning**	Interactive Cytoscape / Vis.js network visualizer
Book Bridge (Shortest Path)	/book_bridge	—	**✓ Path**	**✓ Path**	**✓ Path**	**✓ Path**	Neo4j shortest path traversal between book nodes
Centrality Analytics Lab	/library_analytics	—	**✓ Lab**	**✓ Lab**	**✓ Lab**	**Blocked**	Author Degree, Betweenness & Closeness centrality
Change Password	/change_password	—	✓	✓	✓	✓	Self-service credential updates (8–64 character policy)

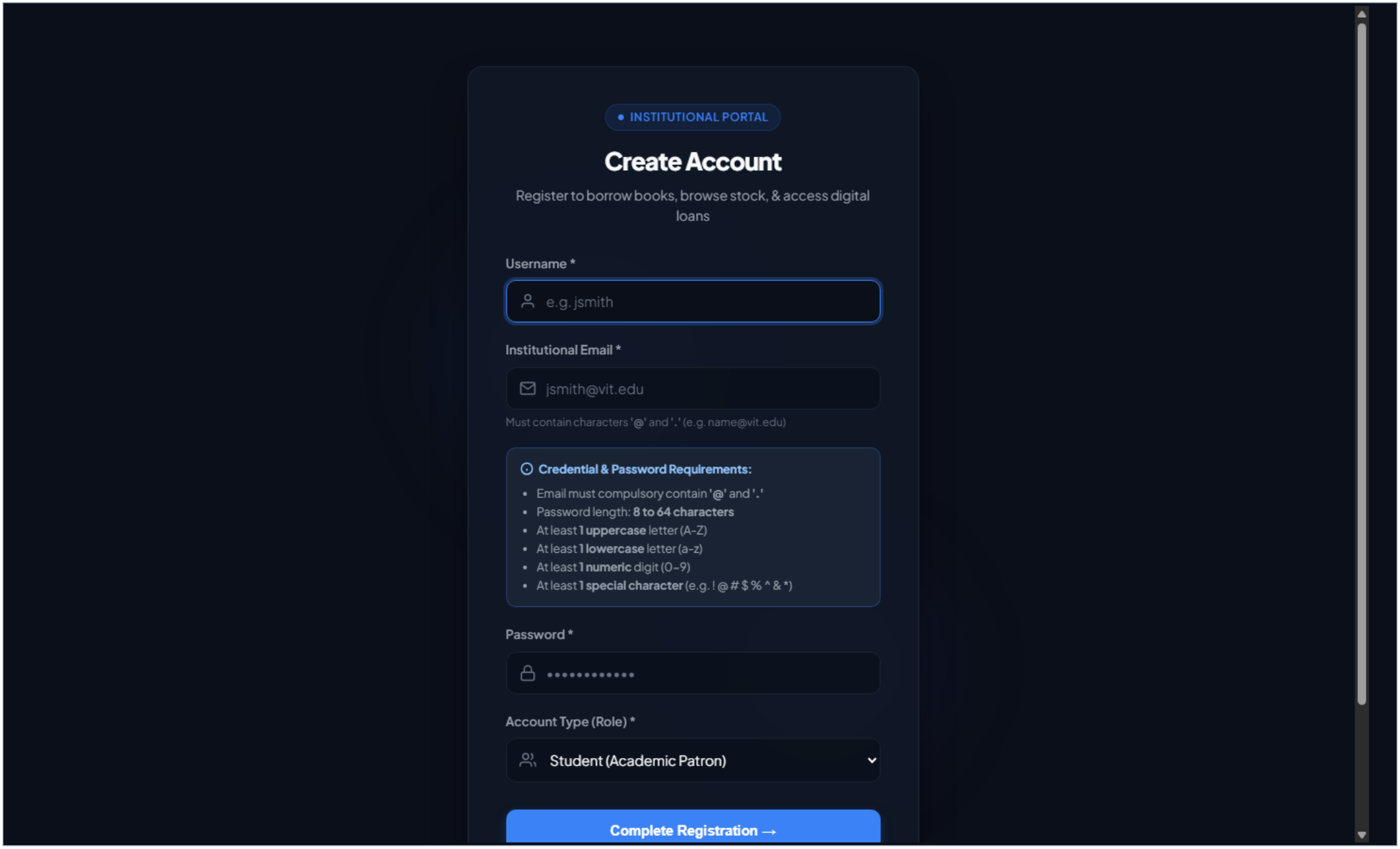
Login Authentication Portal

● PUBLIC ACCESS

Route: http://127.0.0.1:5000/login | **ID:** 01_public_login

VITCC Library UI Audit



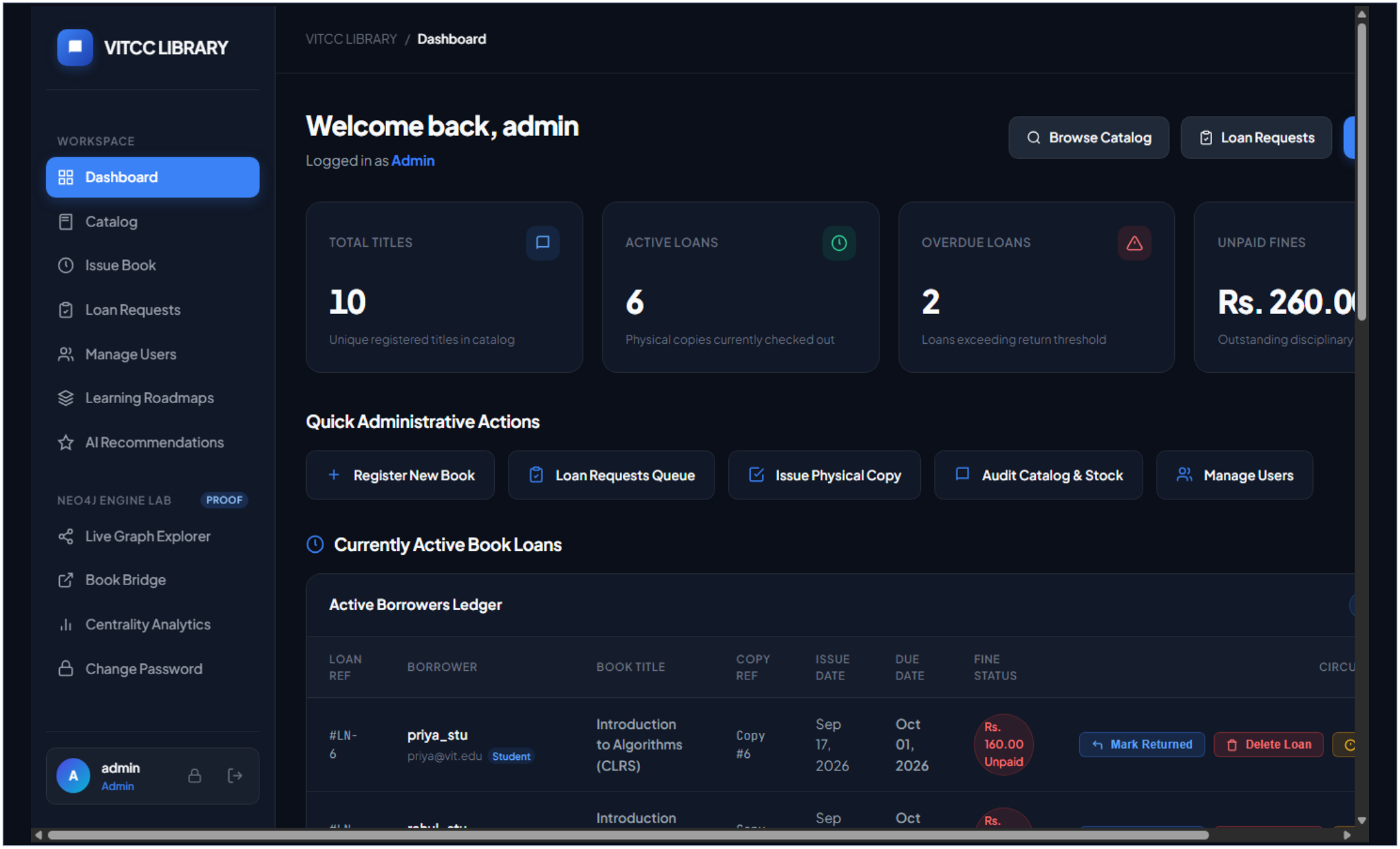

INSTITUTIONAL PORTAL
Create Account
Register to borrow books, browse stock, & access digital loans
Username *
e.g. jsmith
Institutional Email *
jsmith@vit.edu
Must contain characters '@' and '.' (e.g. name@vit.edu)
Credential & Password Requirements:
Email must compulsory contain '@' and '.'
Password length: 8 to 64 characters
At least 1 uppercase letter (A-Z)
At least 1 lowercase letter (a-z)
At least 1 numeric digit (0-9)
At least 1 special character (e.g. ! @ # $ % ^ & *)
Password *
Account Type (Role) *
Student (Academic Patron)
Complete Registration →

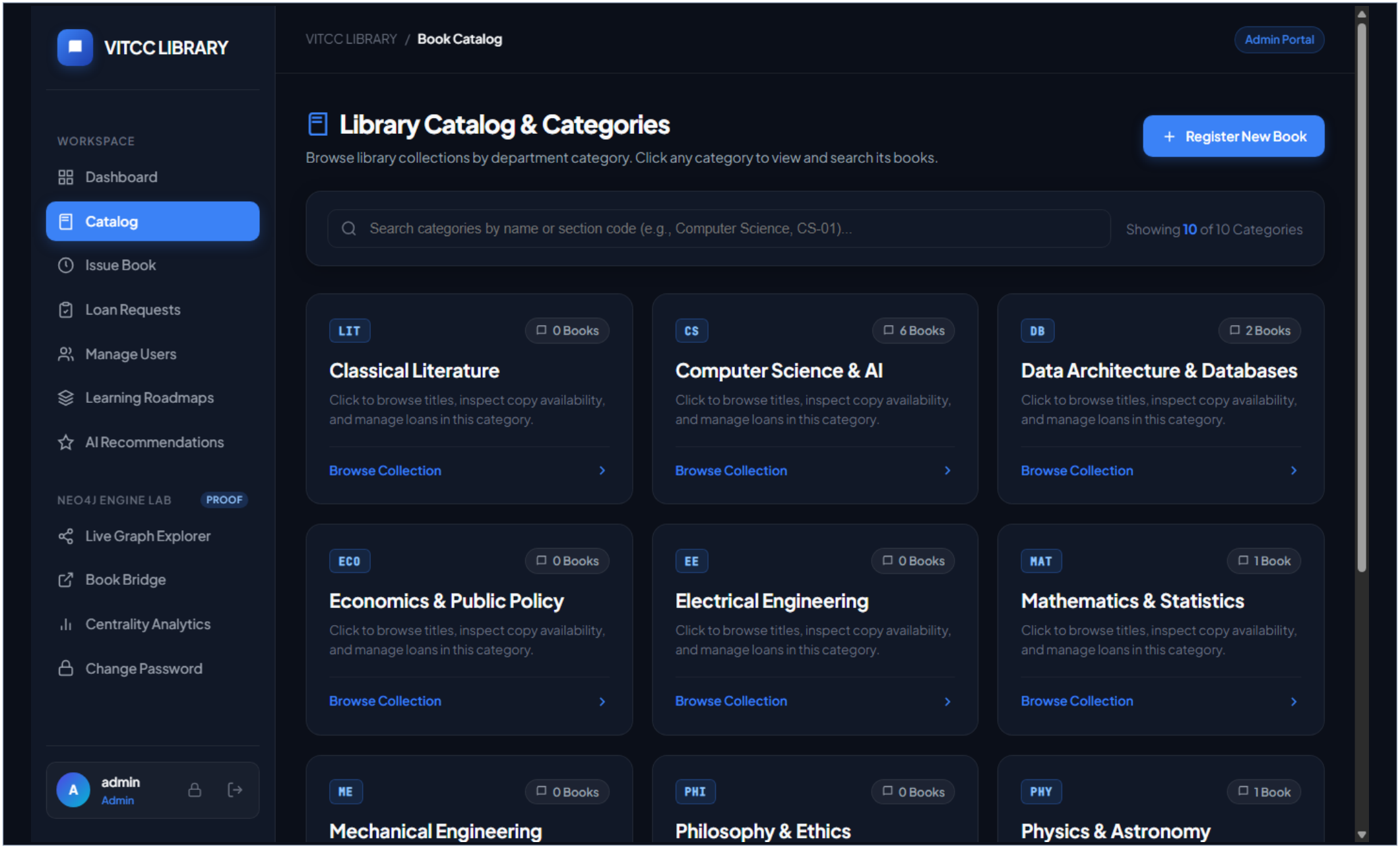
Executive Dashboard & System Metrics

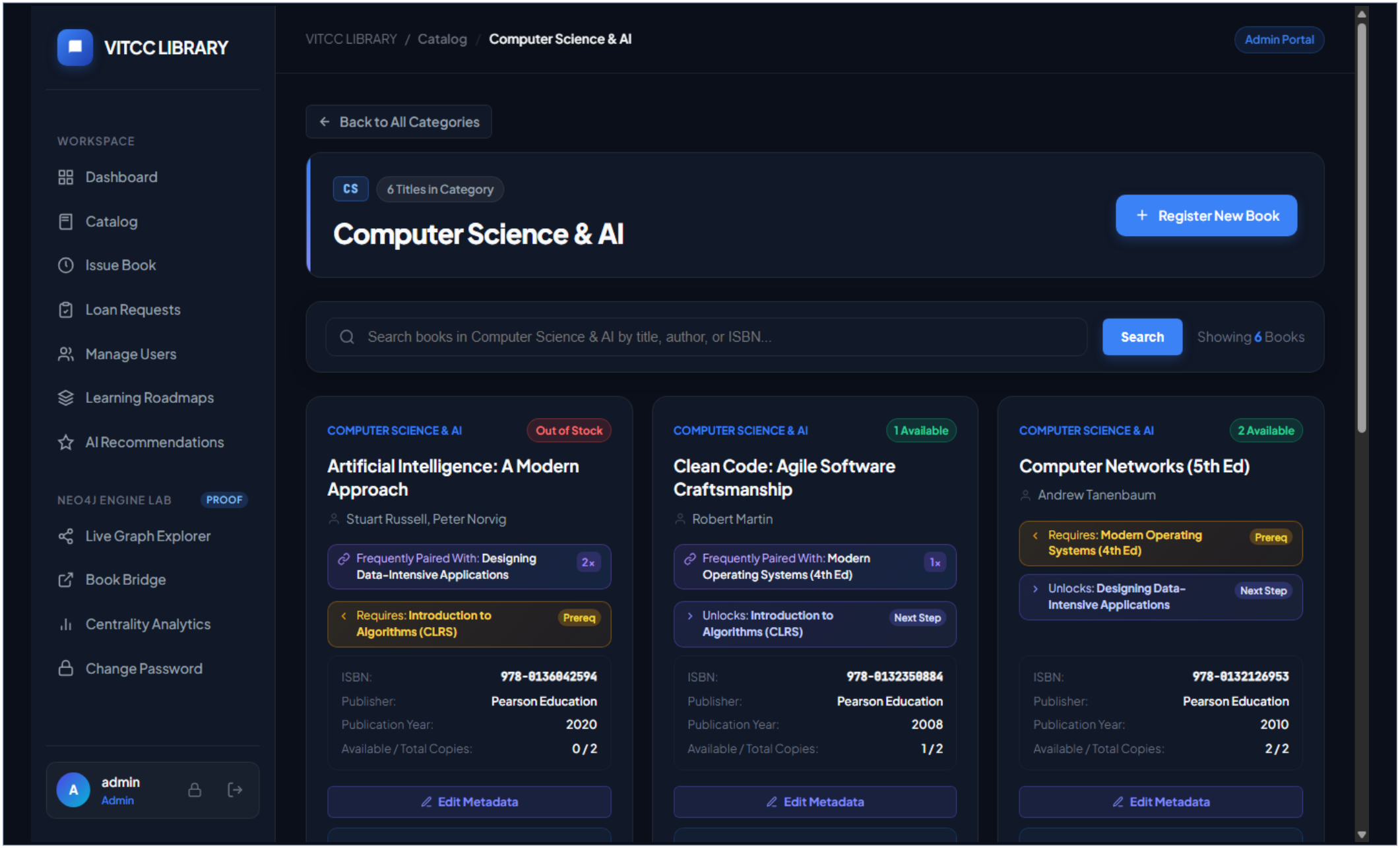
● ADMIN ACCESS

Route: http://127.0.0.1:5000/dashboard | **ID:** 03_admin_dashboard

VITCC Library UI Audit




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VITCC LIBRARY / Book Catalog
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NEO4J ENGINE LAB
PROOF
Live Graph Explorer
Book Bridge
Centrality Analytics
Change Password
admin
Admin
Library Catalog & Categories
Browse library collections by department category. Click any category to view and search its books.
+ Register New Book
Search categories by name or section code (e.g., Computer Science, CS-01)...
Showing 10 of 10 Categories
LIT
0 Books
Classical Literature
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
CS
6 Books
Computer Science & AI
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
DB
2 Books
Data Architecture & Databases
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
ECO
0 Books
Economics & Public Policy
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
EE
0 Books
Electrical Engineering
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
MAT
1 Book
Mathematics & Statistics
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
ME
0 Books
Mechanical Engineering
PHI
0 Books
Philosophy & Ethics
PHY
1 Book
Physics & Astronomy

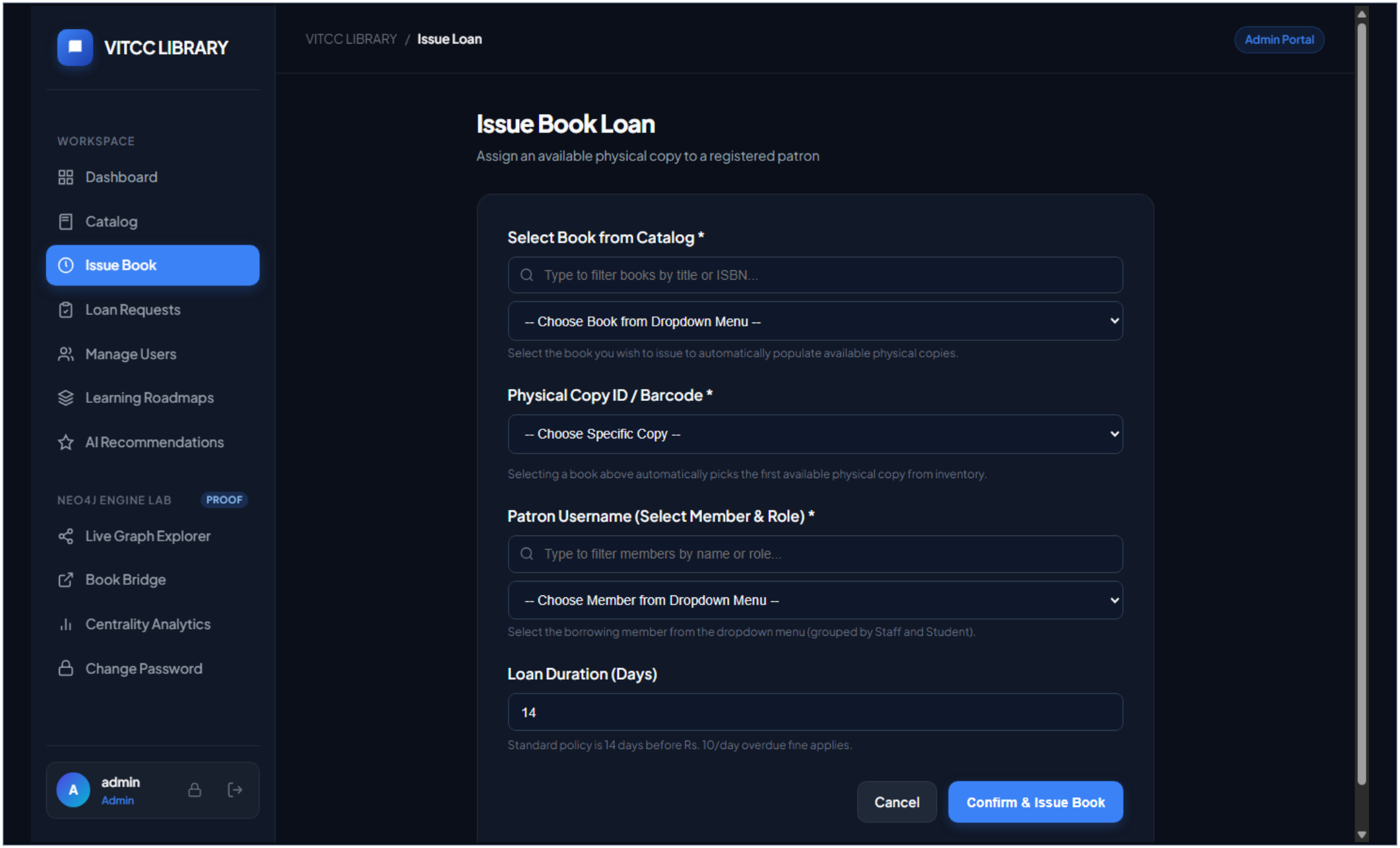

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CS
6 Titles in Category
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COMPUTER SCIENCE & AI
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Artificial Intelligence: A Modern Approach
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Requires: Introduction to Algorithms (CLRS)
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ISBN: 978-0136042594
Publisher: Pearson Education
Publication Year: 2020
Available / Total Copies: 0 / 2
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COMPUTER SCIENCE & AI
1 Available
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COMPUTER SCIENCE & AI
2 Available
Computer Networks (5th Ed)
Andrew Tanenbaum
Requires: Modern Operating Systems (4th Ed)
Prereq
Unlocks: Designing Data-Intensive Applications
Next Step
ISBN: 978-0132126953
Publisher: Pearson Education
Publication Year: 2010
Available / Total Copies: 2 / 2
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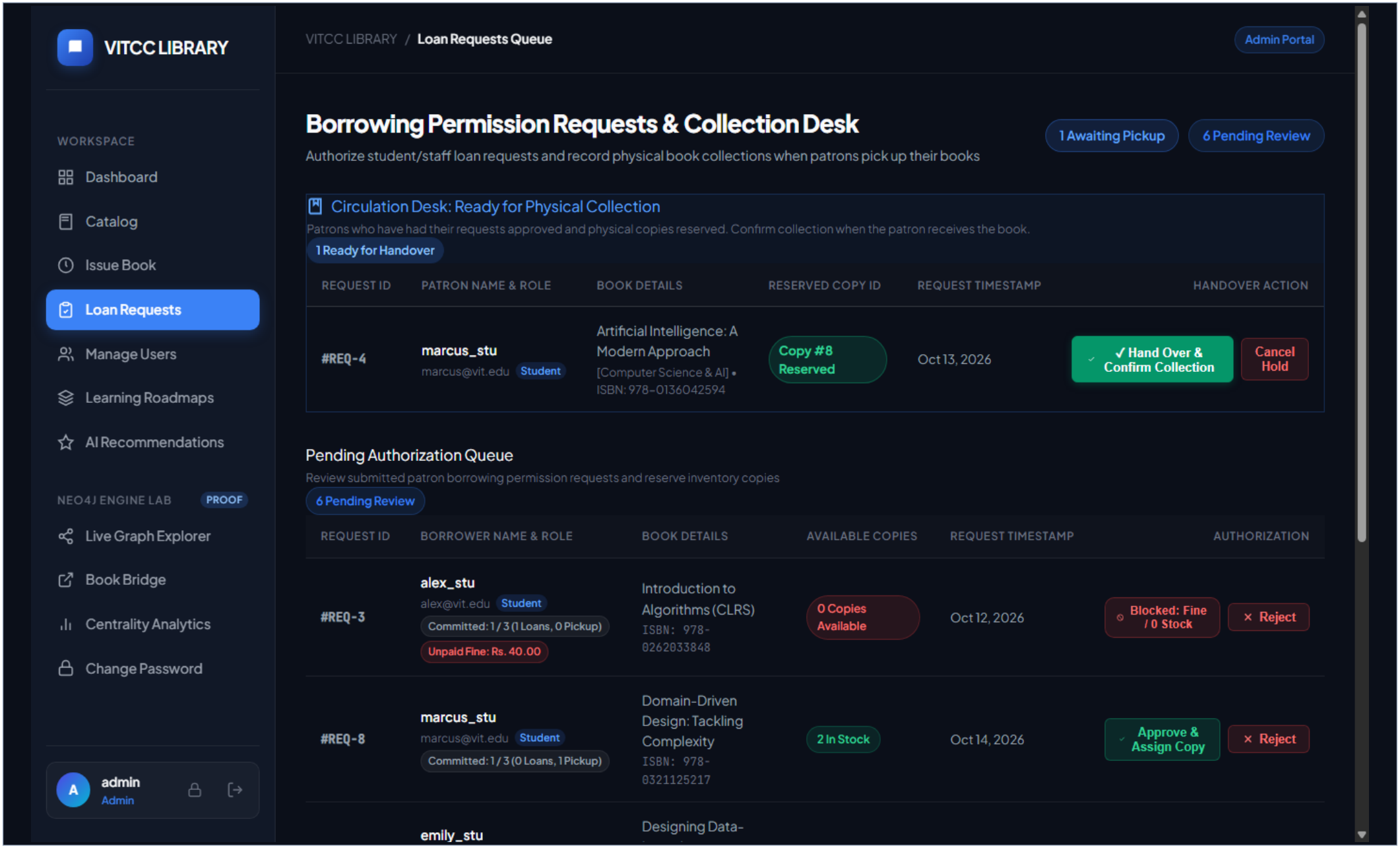
Circulation Desk - Direct Book Issuance

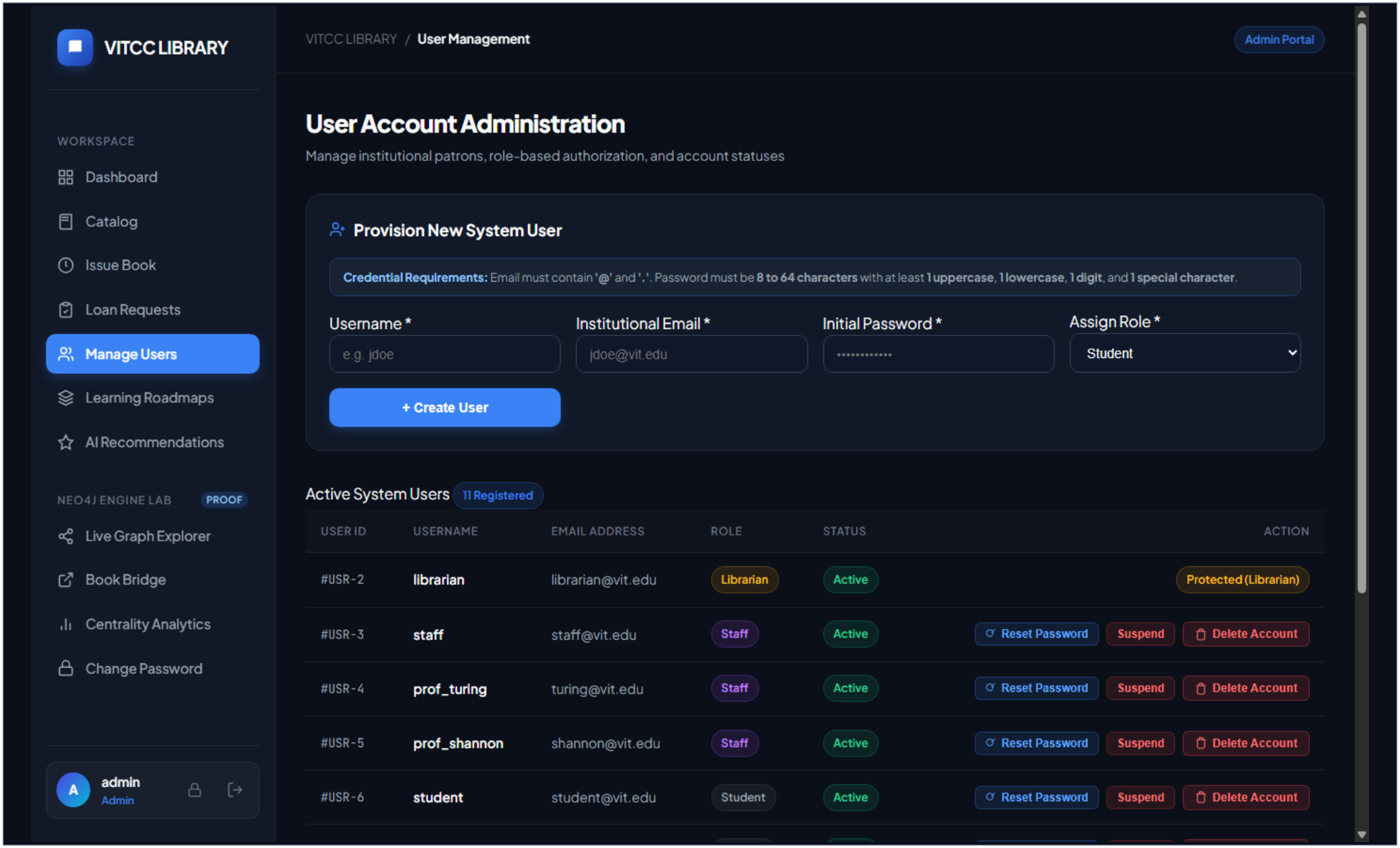
● ADMIN ACCESS

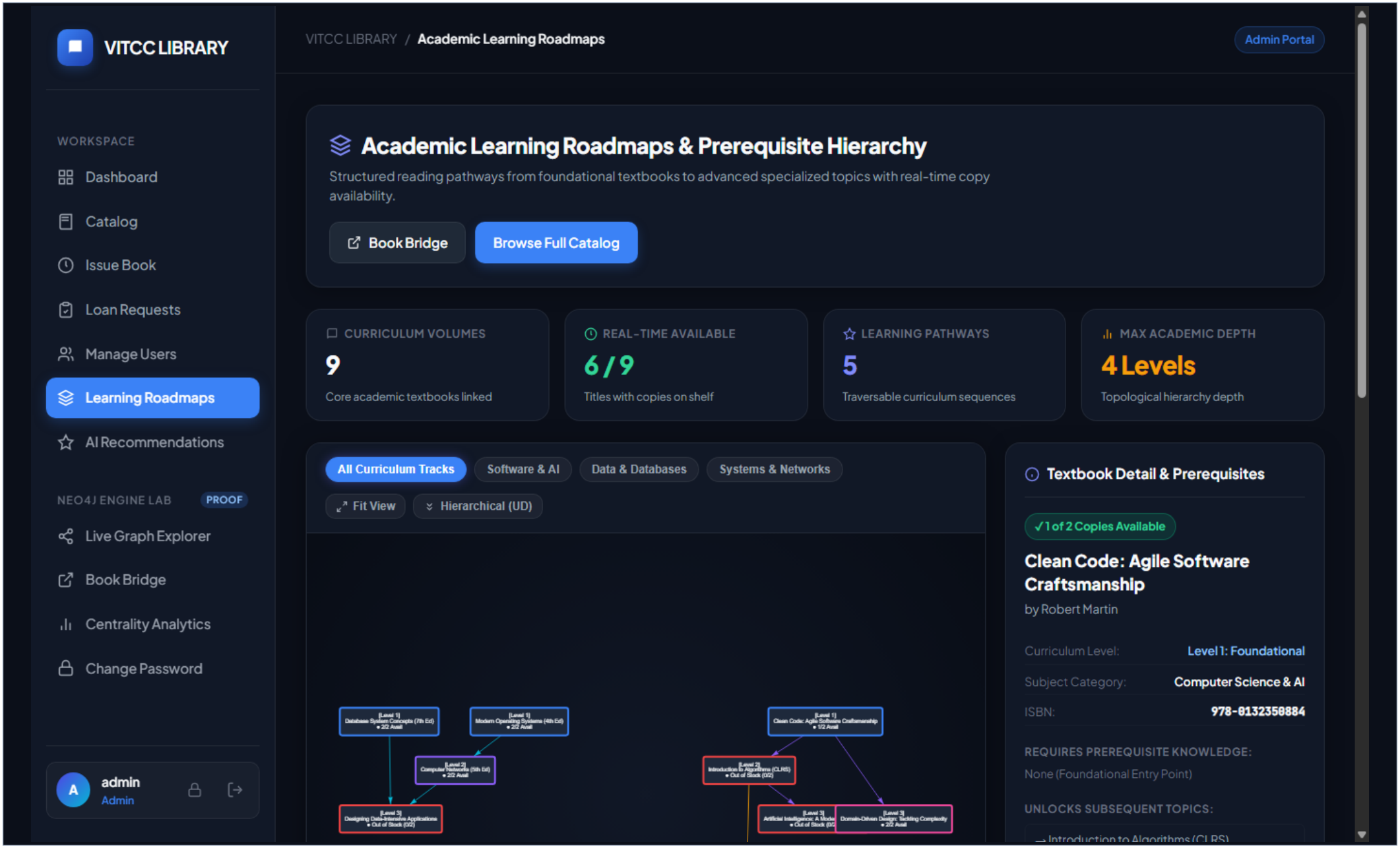
Route: http://127.0.0.1:5000/issue_loan | **ID:** 06_admin_issue_loan

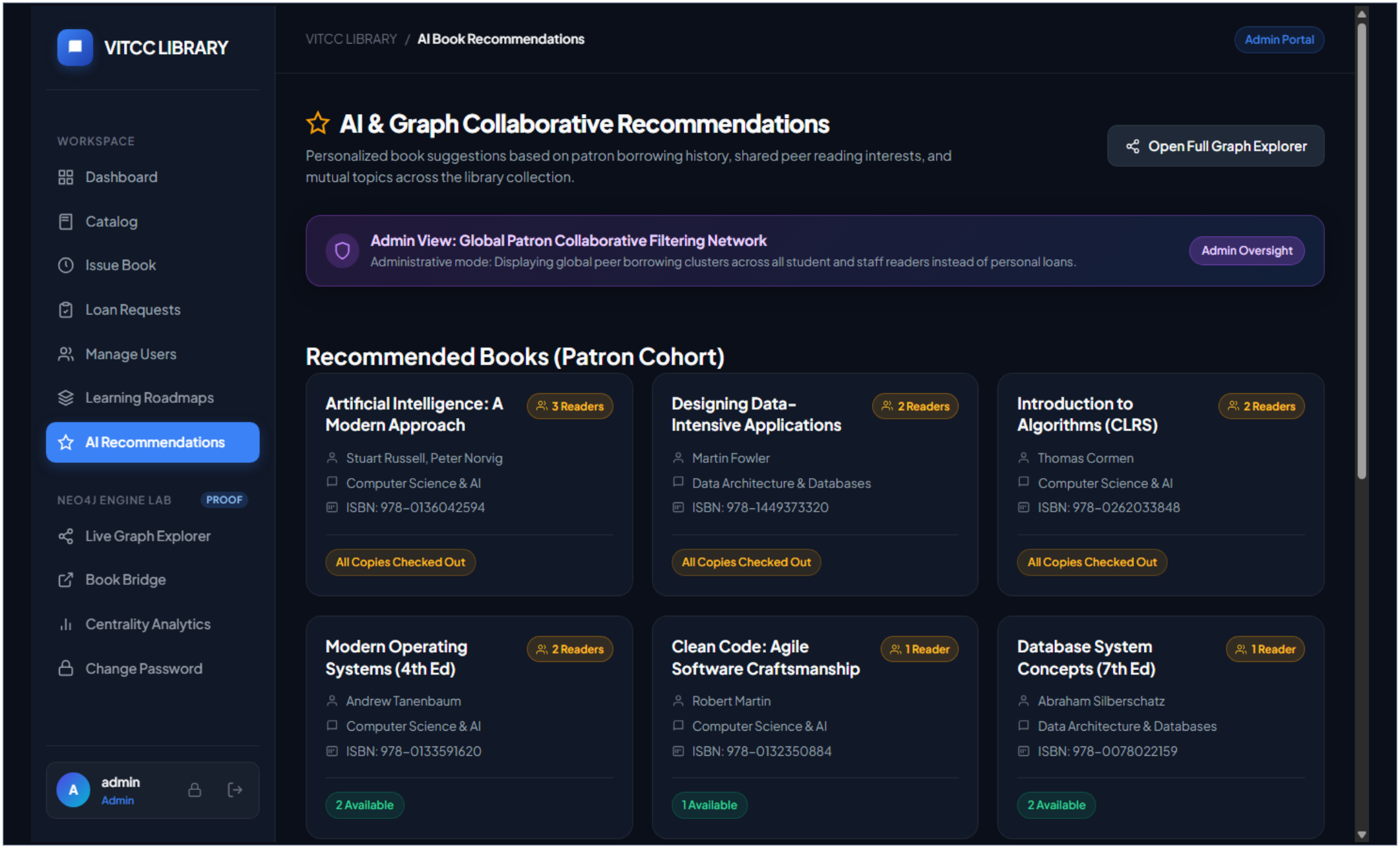
VITCC Library UI Audit




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VITCC LIBRARY / Loan Requests Queue
Admin Portal
Borrowing Permission Requests & Collection Desk
Authorize student/staff loan requests and record physical book collections when patrons pick up their books
1 Awaiting Pickup
6 Pending Review
Circulation Desk: Ready for Physical Collection
Patrons who have had their requests approved and physical copies reserved. Confirm collection when the patron receives the book.
1 Ready for Handover
REQUEST ID
PATRON NAME & ROLE
BOOK DETAILS
RESERVED COPY ID
REQUEST TIMESTAMP
HANDOVER ACTION
#REQ-4
marcus_stu
marcus@vit.edu Student
Artificial Intelligence: A Modern Approach
[Computer Science & AI] • ISBN: 978-0136042594
Copy #8 Reserved
Oct 13, 2026
✓ Hand Over & Confirm Collection
Cancel Hold
Pending Authorization Queue
Review submitted patron borrowing permission requests and reserve inventory copies
6 Pending Review
REQUEST ID
BORROWER NAME & ROLE
BOOK DETAILS
AVAILABLE COPIES
REQUEST TIMESTAMP
AUTHORIZATION
#REQ-3
alex_stu
alex@vit.edu Student
Committed: 1 / 3 (1 Loans, 0 Pickup)
Unpaid Fine: Rs. 40.00
Introduction to Algorithms (CLRS)
ISBN: 978-0262033848
0 Copies Available
Oct 12, 2026
Blocked: Fine / 0 Stock
× Reject
#REQ-8
marcus_stu
marcus@vit.edu Student
Committed: 1 / 3 (0 Loans, 1 Pickup)
Domain-Driven Design: Tackling Complexity
ISBN: 978-0321125217
2 In Stock
Oct 14, 2026
Approve & Assign Copy
× Reject
emily_stu
Designing Data-


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admin
Admin
User Account Administration
Manage institutional patrons, role-based authorization, and account statuses
Provision New System User
Credential Requirements: Email must contain '@' and '.'. Password must be 8 to 64 characters with at least 1 uppercase, 1 lowercase, 1 digit, and 1 special character.
Username *
e.g. jdoe
Institutional Email *
jdoe@vit.edu
Initial Password *
Assign Role *
Student
+ Create User
Active System Users
11 Registered
USER ID
USERNAME
EMAIL ADDRESS
ROLE
STATUS
ACTION
#USR-2
librarian
librarian@vit.edu
Librarian
Active
Protected (Librarian)
#USR-3
staff
staff@vit.edu
Staff
Active
Reset Password
Suspend
Delete Account
#USR-4
prof_turing
turing@vit.edu
Staff
Active
Reset Password
Suspend
Delete Account
#USR-5
prof_shannon
shannon@vit.edu
Staff
Active
Reset Password
Suspend
Delete Account
#USR-6
student
student@vit.edu
Student
Active
Reset Password
Suspend
Delete Account


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Academic Learning Roadmaps & Prerequisite Hierarchy
Structured reading pathways from foundational textbooks to advanced specialized topics with real-time copy availability.
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5
Traversable curriculum sequences
MAX ACADEMIC DEPTH
4 Levels
Topological hierarchy depth
All Curriculum Tracks
Software & AI
Data & Databases
Systems & Networks
Fit View
Hierarchical (UD)
Textbook Detail & Prerequisites
✓1 of 2 Copies Available
Clean Code: Agile Software Craftsmanship
by Robert Martin
Curriculum Level:
Level 1: Foundational
Subject Category:
Computer Science & AI
ISBN:
978-0132350884
REQUIRES PREREQUISITE KNOWLEDGE:
None (Foundational Entry Point)
UNLOCKS SUBSEQUENT TOPICS:

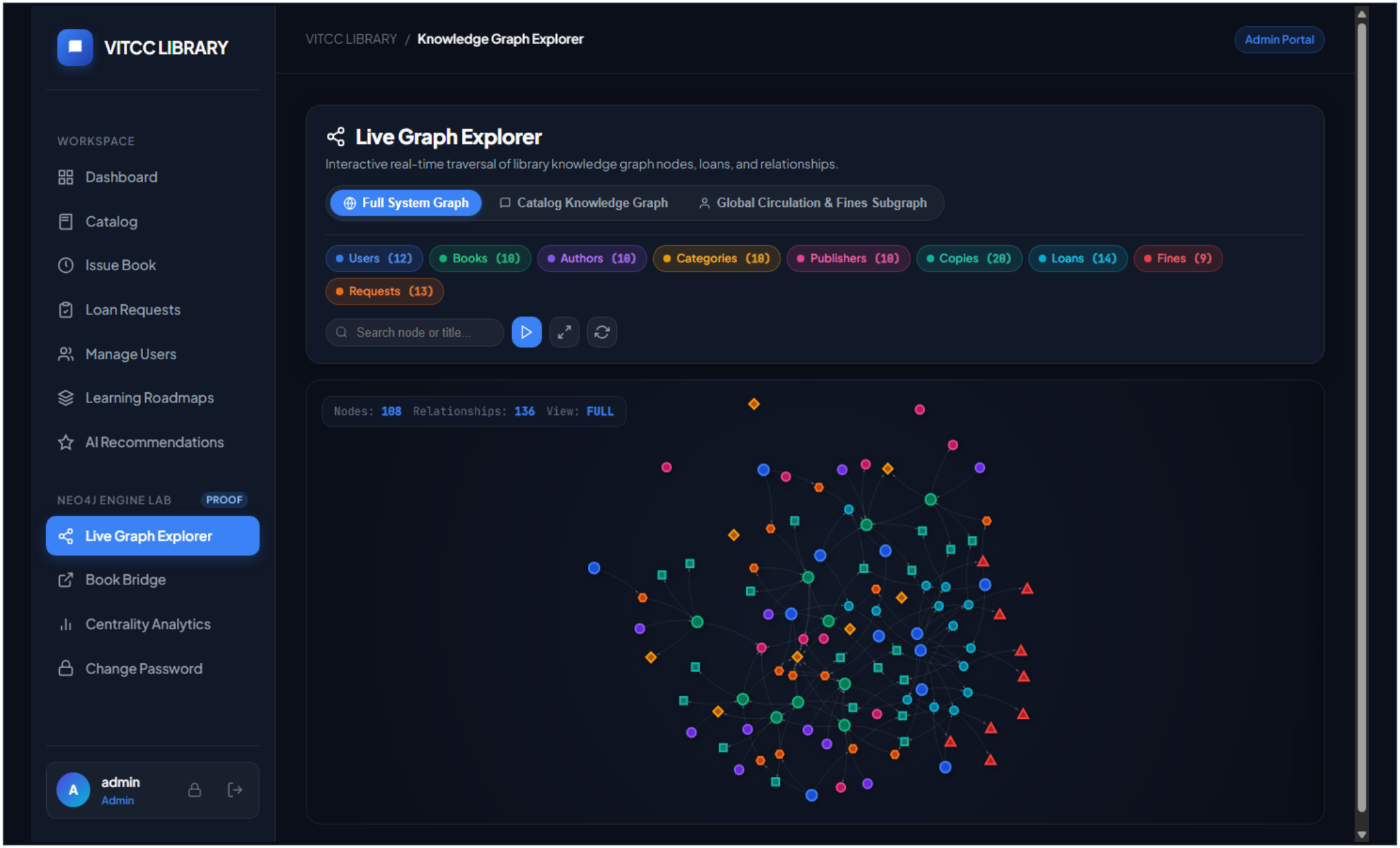

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Admin
AI & Graph Collaborative Recommendations
Personalized book suggestions based on patron borrowing history, shared peer reading interests, and mutual topics across the library collection.
Open Full Graph Explorer
Admin View: Global Patron Collaborative Filtering Network
Administrative mode: Displaying global peer borrowing clusters across all student and staff readers instead of personal loans.
Admin Oversight
Recommended Books (Patron Cohort)
Artificial Intelligence: A Modern Approach
3 Readers
Stuart Russell, Peter Norvig
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All Copies Checked Out
Designing Data-Intensive Applications
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Database System Concepts (7th Ed)
1 Reader
Abraham Silberschatz
Data Architecture & Databases
ISBN: 978-0078022159
2 Available

Global Knowledge Graph Explorer

● ADMIN ACCESS

Route: http://127.0.0.1:5000/graph_visualizer | **ID:** 11_admin_graph_visualizer

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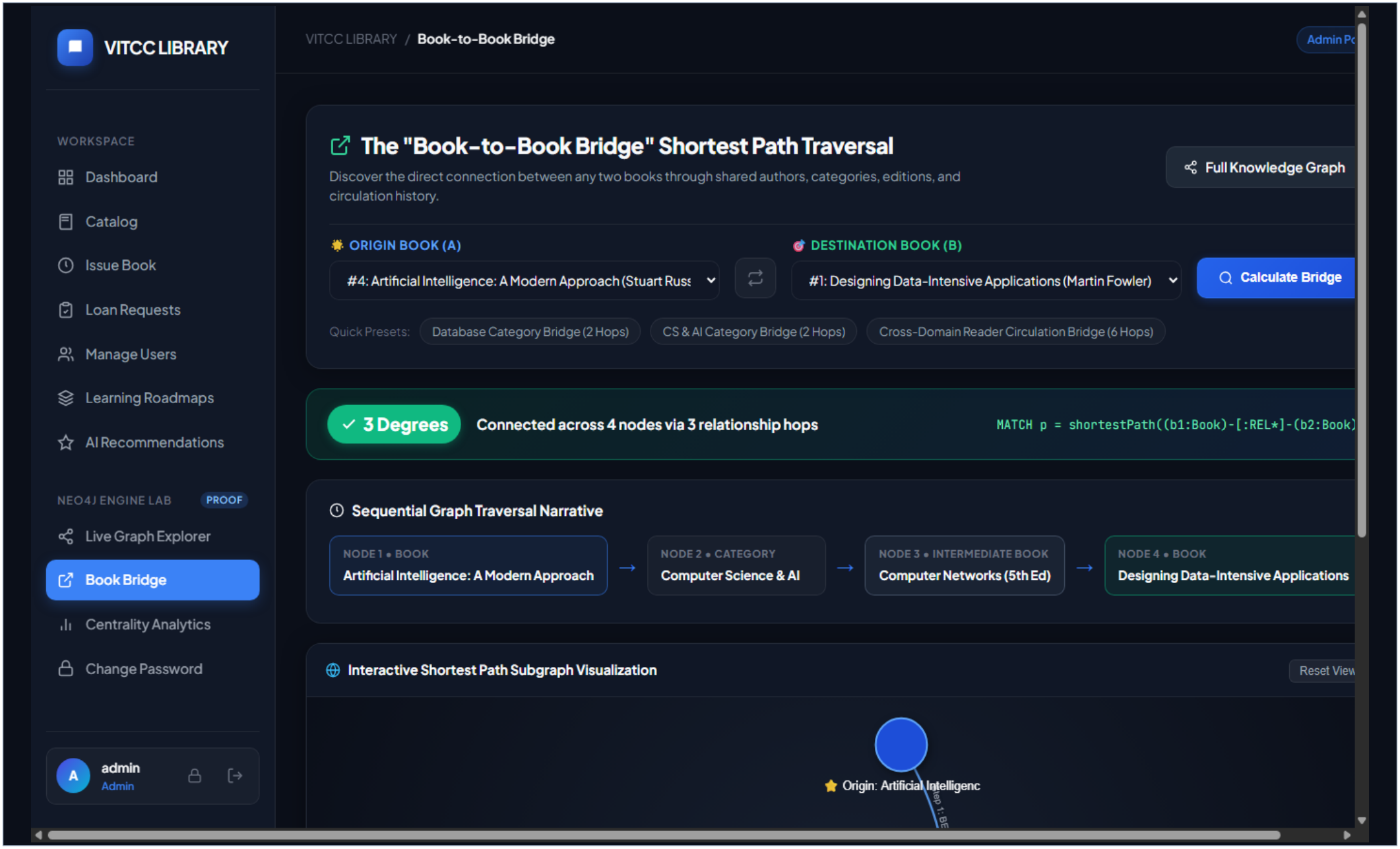


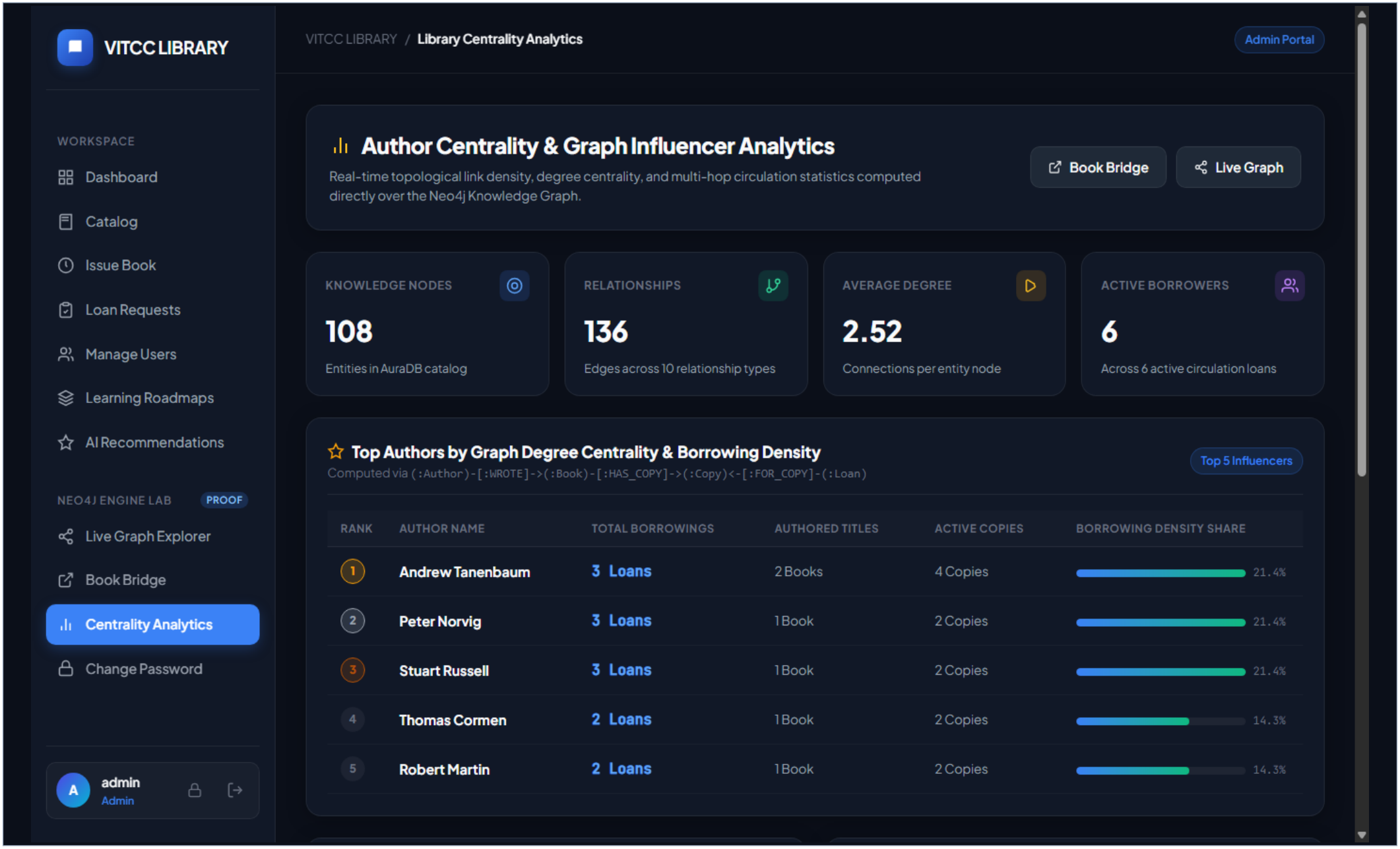
Book Bridge - Shortest Path Separation

● ADMIN ACCESS

Route: http://127.0.0.1:5000/book_bridge | **ID:** 12_admin_book_bridge

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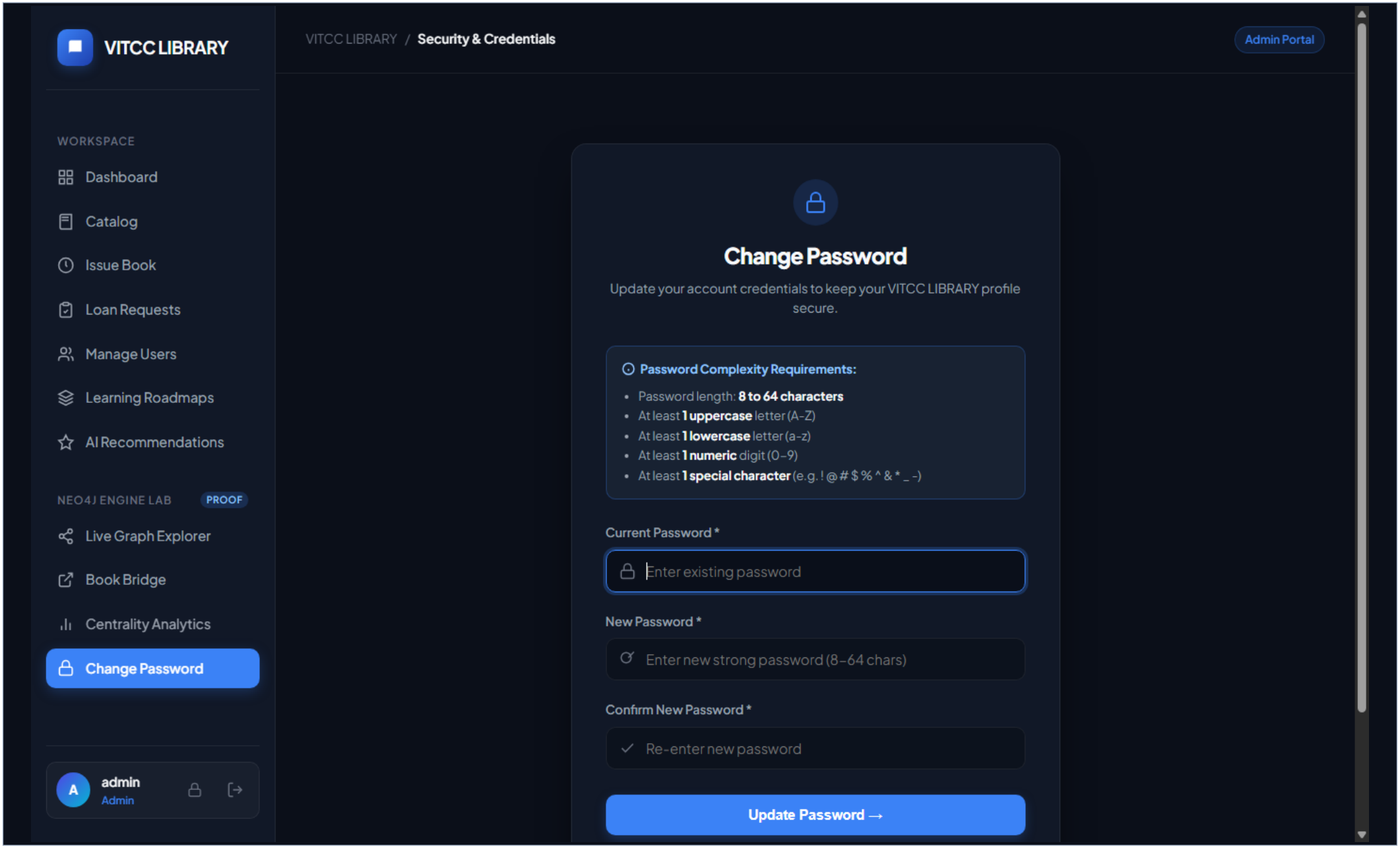

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AI Recommendations
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PROOF
Live Graph Explorer
Book Bridge
Centrality Analytics
Change Password
admin
Admin
Author Centrality & Graph Influencer Analytics
Real-time topological link density, degree centrality, and multi-hop circulation statistics computed directly over the Neo4j Knowledge Graph.
Book Bridge
Live Graph
KNOWLEDGE NODES
108
Entities in AuraDB catalog
RELATIONSHIPS
136
Edges across 10 relationship types
AVERAGE DEGREE
2.52
Connections per entity node
ACTIVE BORROWERS
6
Across 6 active circulation loans
Top Authors by Graph Degree Centrality & Borrowing Density
Computed via (:Author)-[:WROTE]->(:Book)-[:HAS_COPY]->(:Copy)<-[:FOR_COPY]-(:Loan)
Top 5 Influencers
RANK AUTHOR NAME TOTAL BORROWINGS AUTHORED TITLES ACTIVE COPIES BORROWING DENSITY SHARE
1 Andrew Tanenbaum 3 Loans 2 Books 4 Copies 21.4%
2 Peter Norvig 3 Loans 1 Book 2 Copies 21.4%
3 Stuart Russell 3 Loans 1 Book 2 Copies 21.4%
4 Thomas Cormen 2 Loans 1 Book 2 Copies 14.3%
5 Robert Martin 2 Loans 1 Book 2 Copies 14.3%

Security Credentials & Password Management

● ADMIN ACCESS

Route: http://127.0.0.1:5000/change_password | **ID:** 14_admin_change_password

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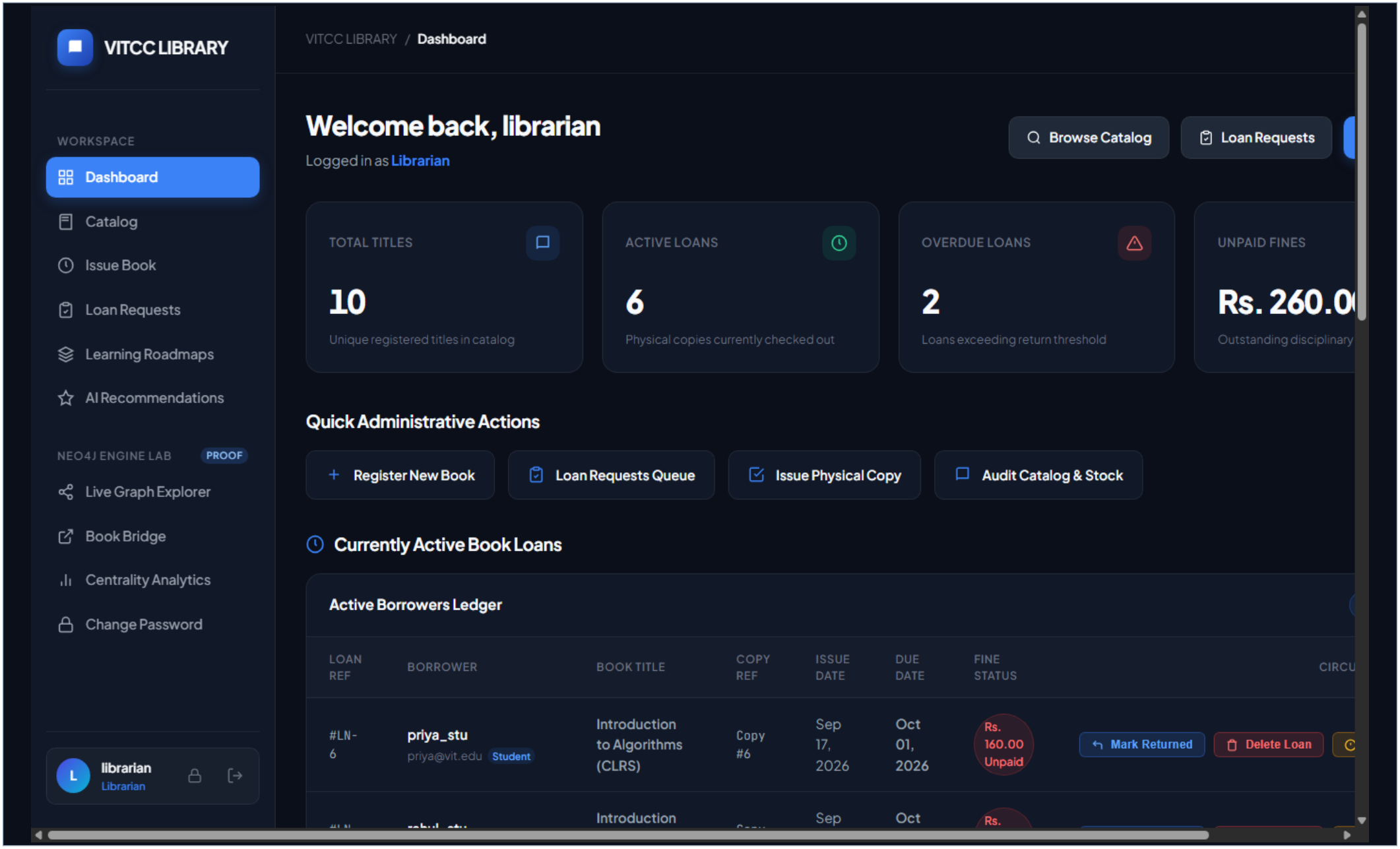


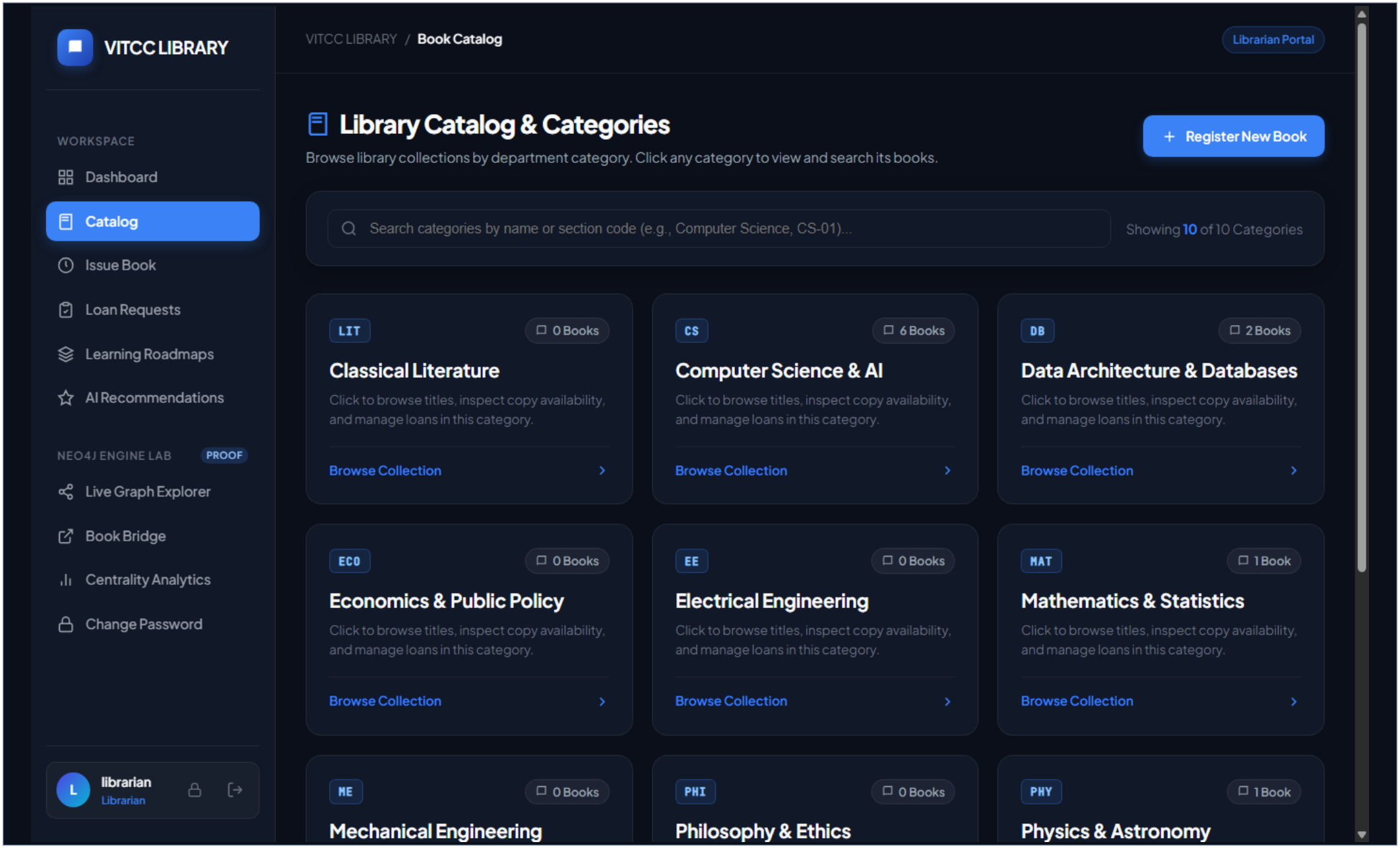
Circulation Desk Dashboard

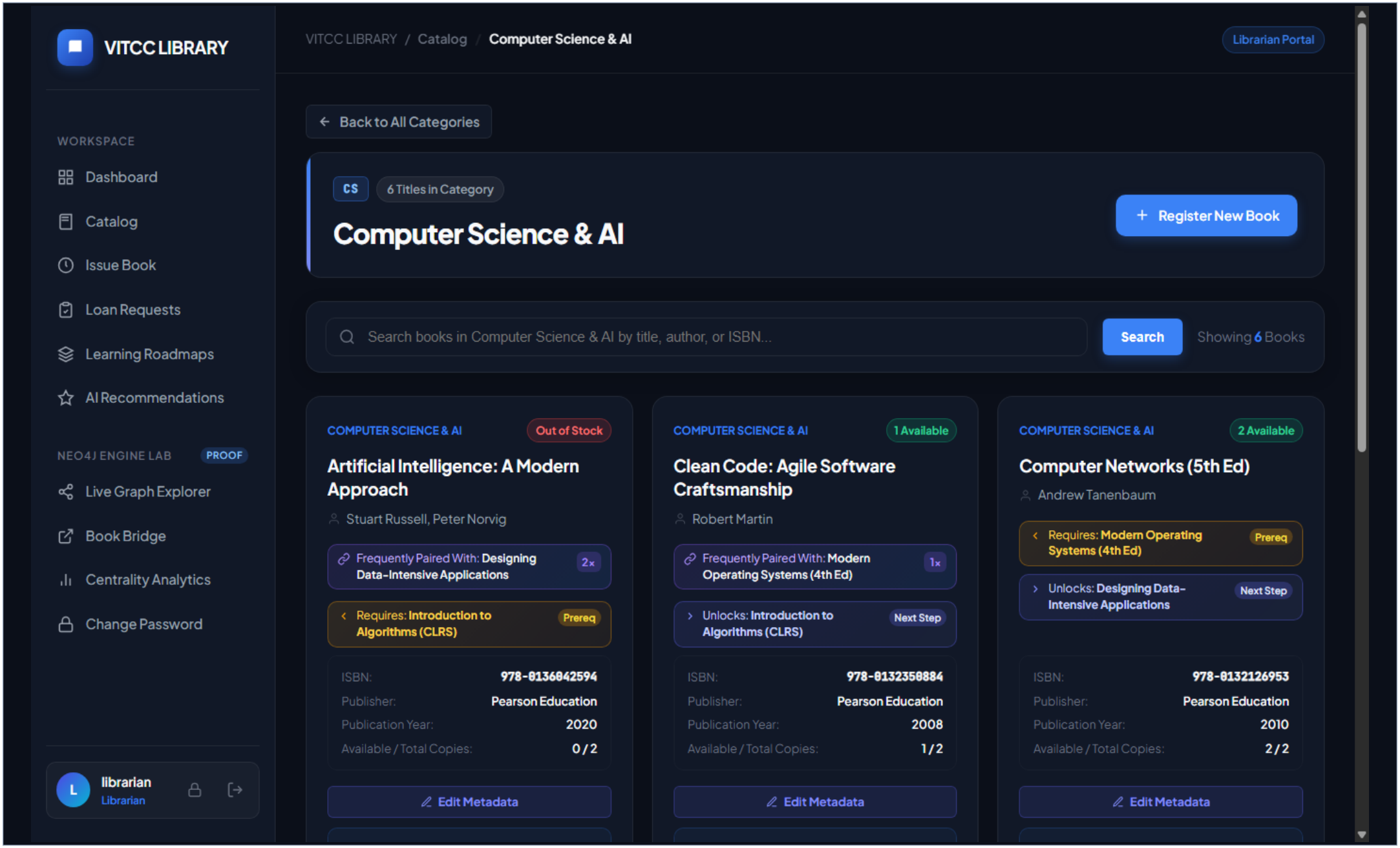
● LIBRARIAN ACCESS

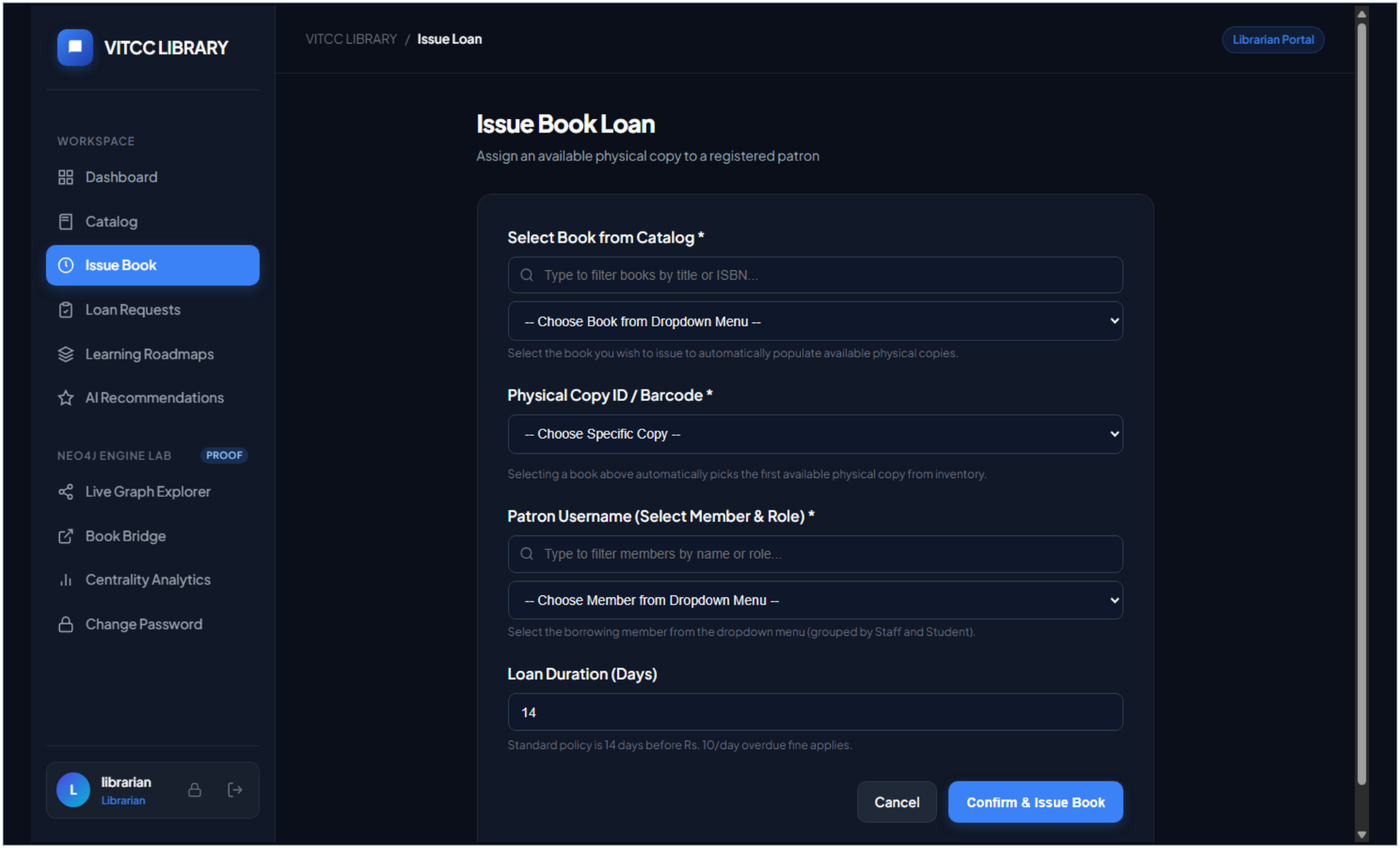
Route: http://127.0.0.1:5000/dashboard | **ID:** 15_librarian_dashboard

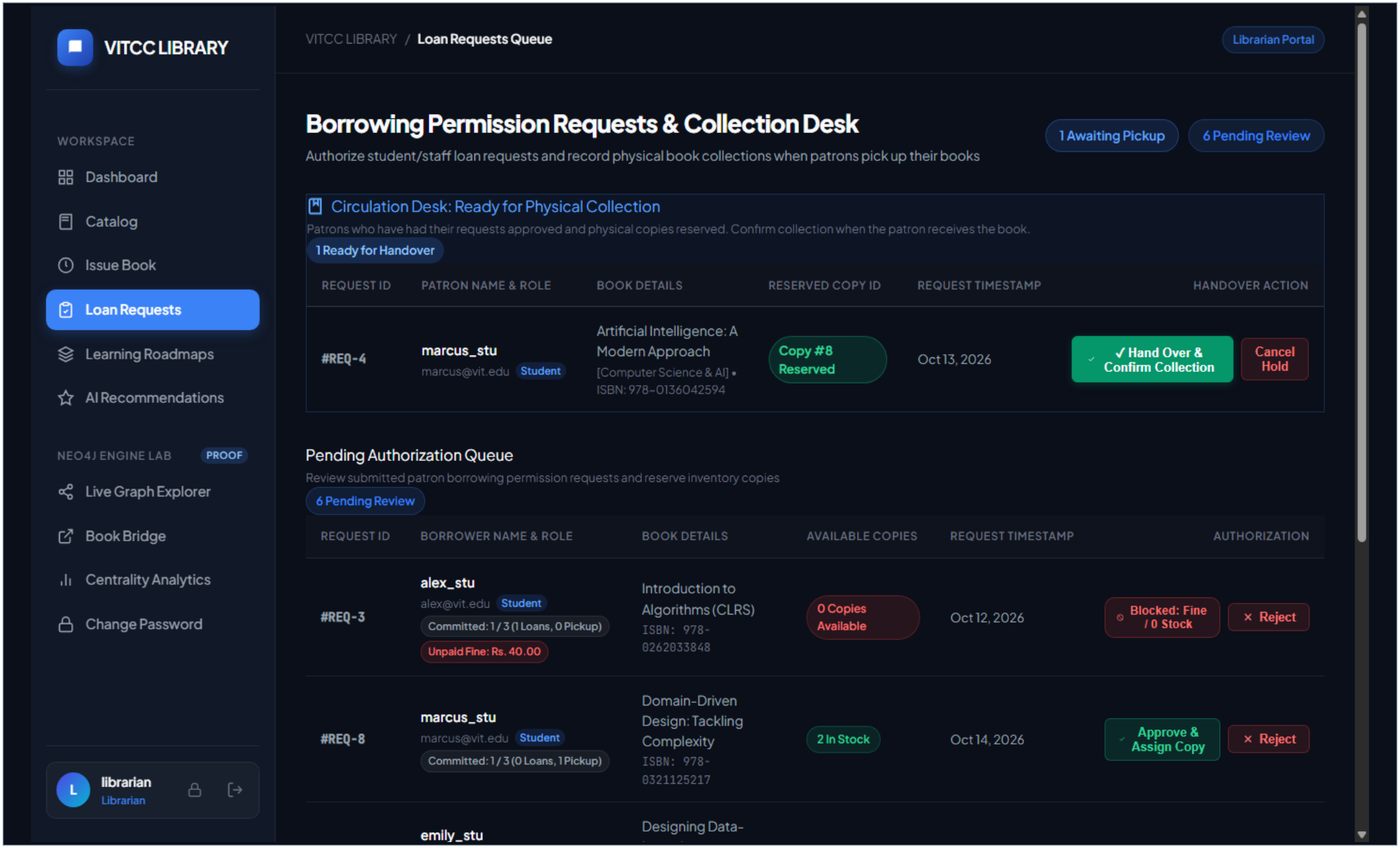
VITCC Library UI Audit

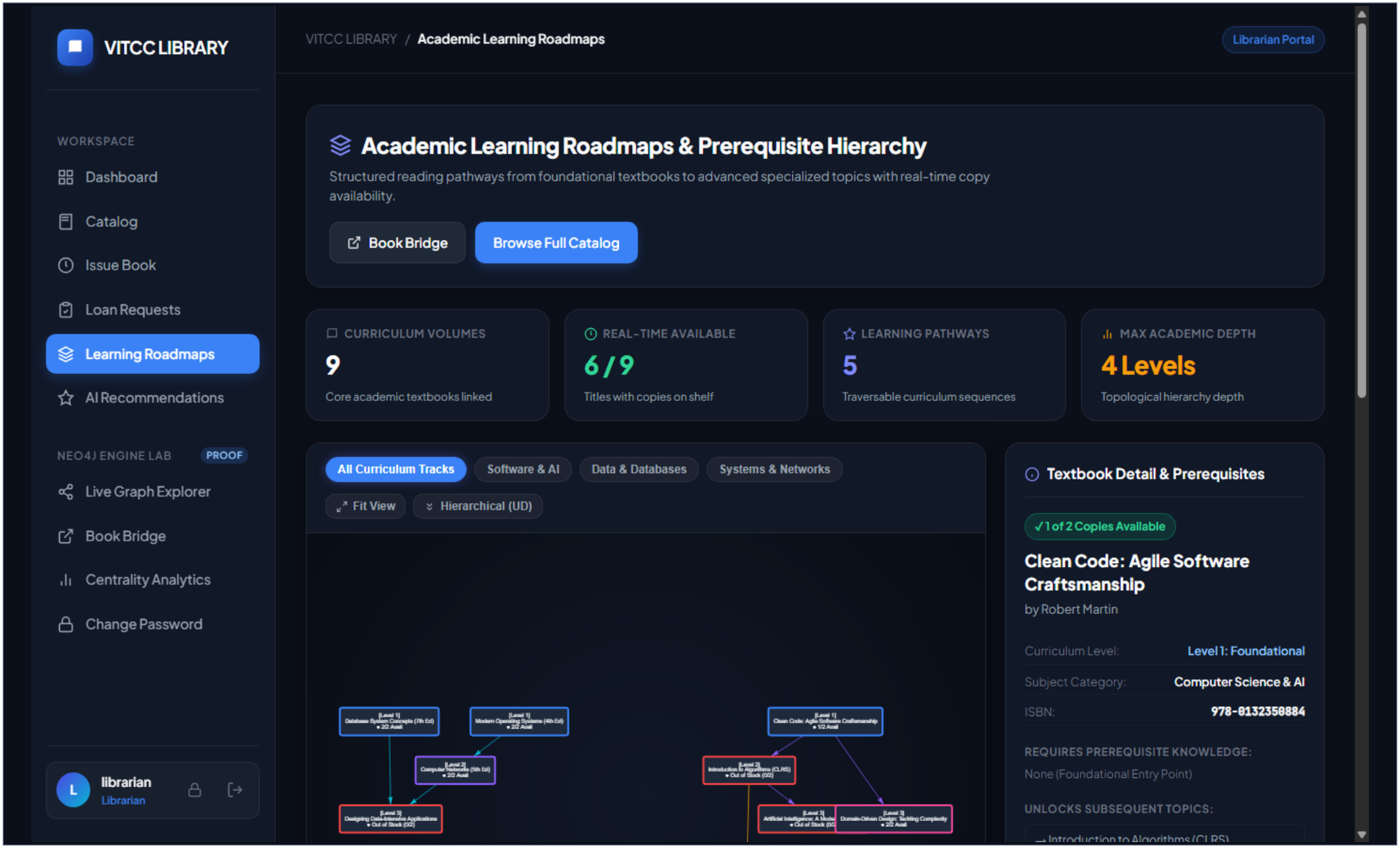


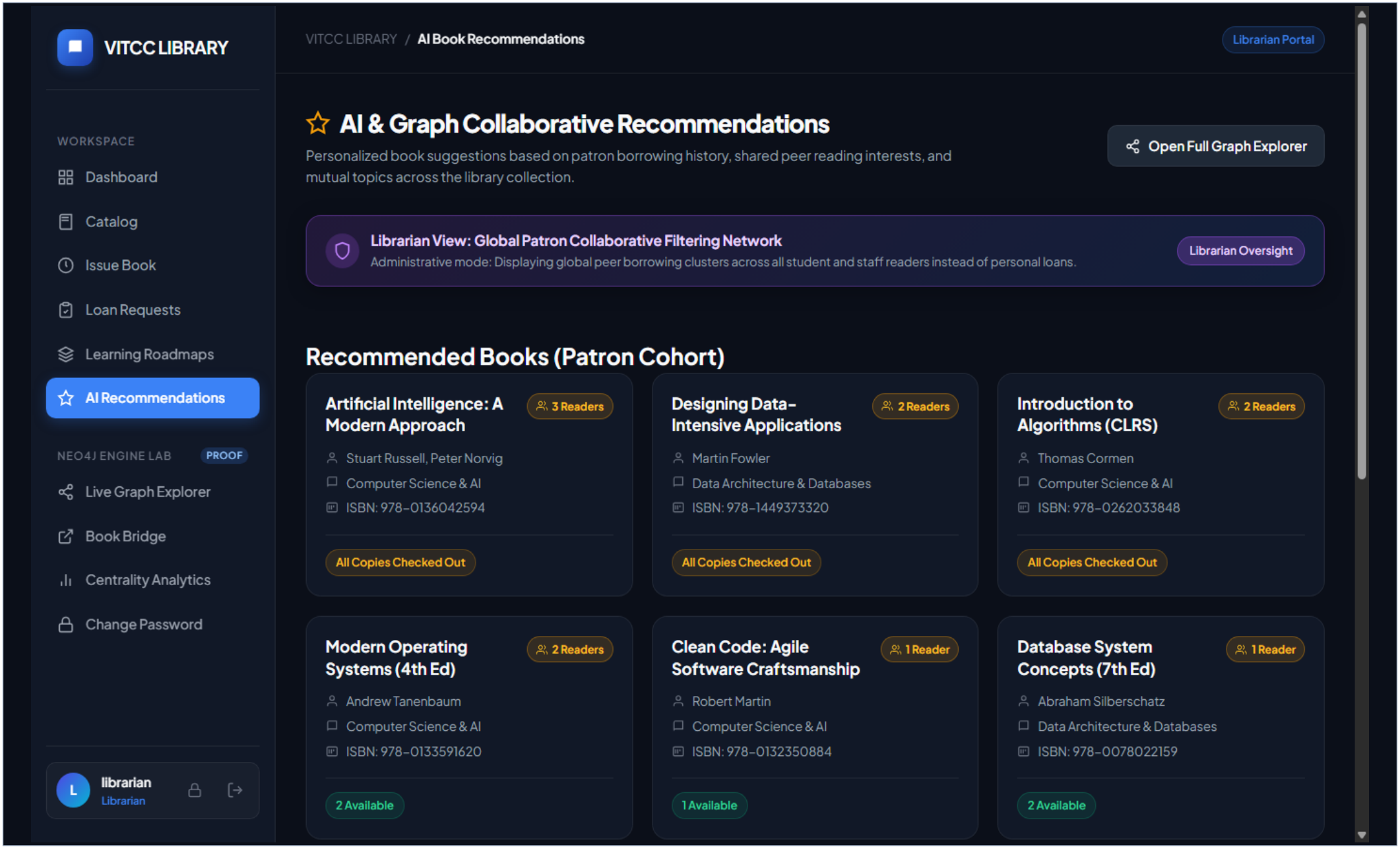

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Change Password
librarian
Librarian
Library Catalog & Categories
Browse library collections by department category. Click any category to view and search its books.
+ Register New Book
Search categories by name or section code (e.g., Computer Science, CS-01)...
Showing 10 of 10 Categories
LIT
0 Books
Classical Literature
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
CS
6 Books
Computer Science & AI
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
DB
2 Books
Data Architecture & Databases
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
ECO
0 Books
Economics & Public Policy
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
EE
0 Books
Electrical Engineering
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
MAT
1 Book
Mathematics & Statistics
Click to browse titles, inspect copy availability, and manage loans in this category.
Browse Collection
ME
0 Books
Mechanical Engineering
PHI
0 Books
Philosophy & Ethics
PHY
1 Book
Physics & Astronomy

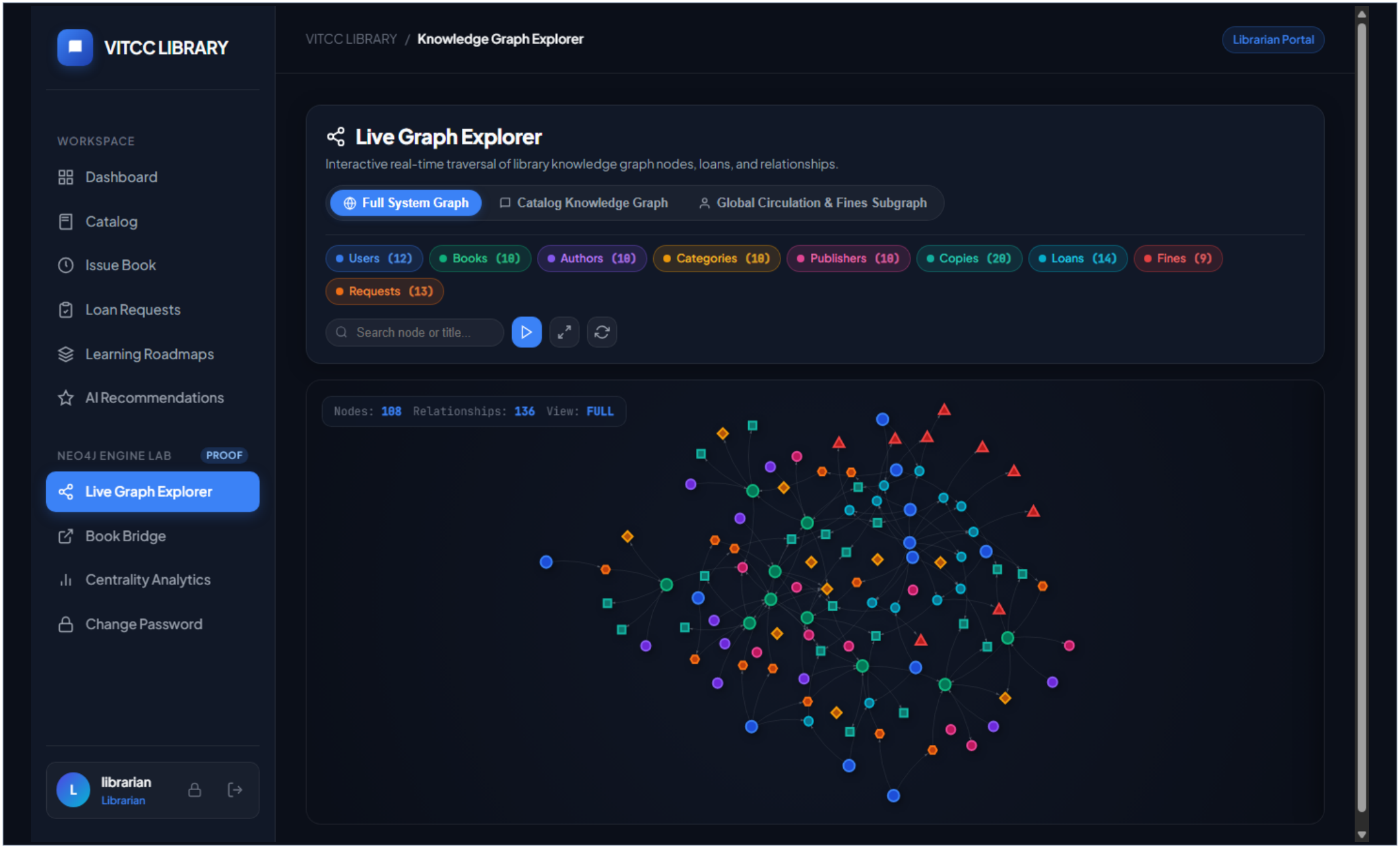

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librarian
Librarian
Back to All Categories
CS
6 Titles in Category
Computer Science & AI
Register New Book
Search books in Computer Science & AI by title, author, or ISBN...
Search
Showing 6 Books
COMPUTER SCIENCE & AI
Out of Stock
Artificial Intelligence: A Modern Approach
Stuart Russell, Peter Norvig
Frequently Paired With: Designing Data-Intensive Applications
2×
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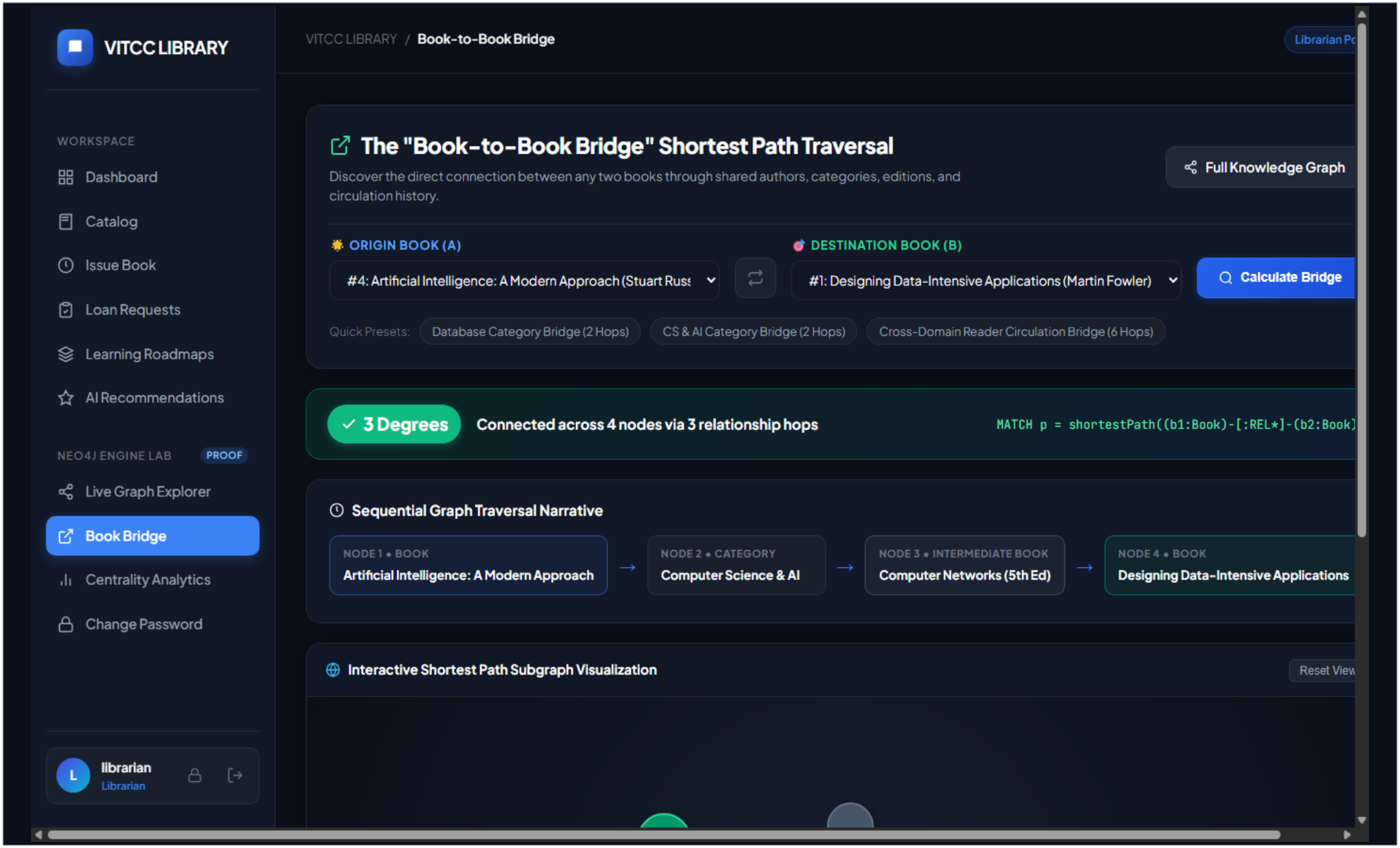

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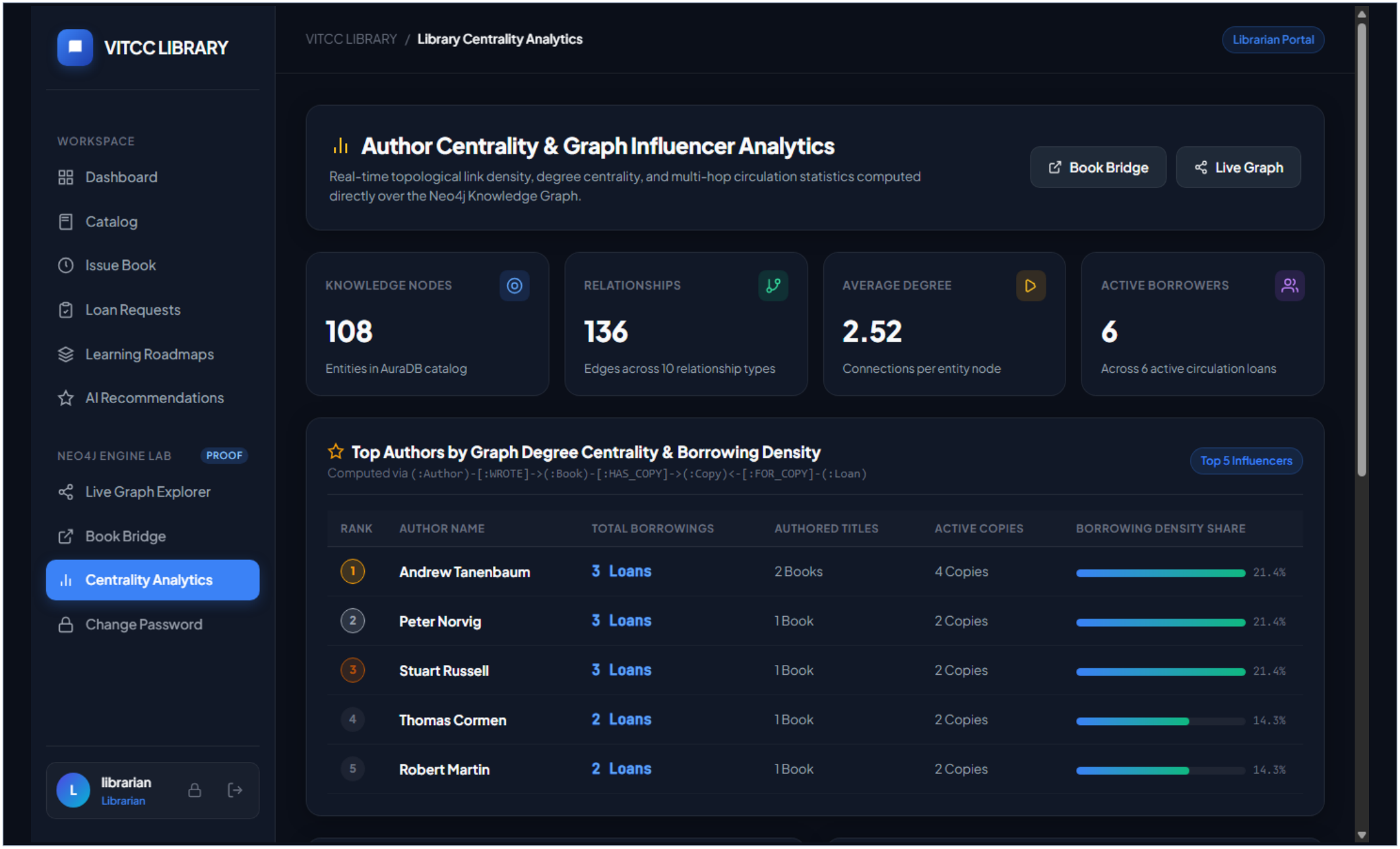
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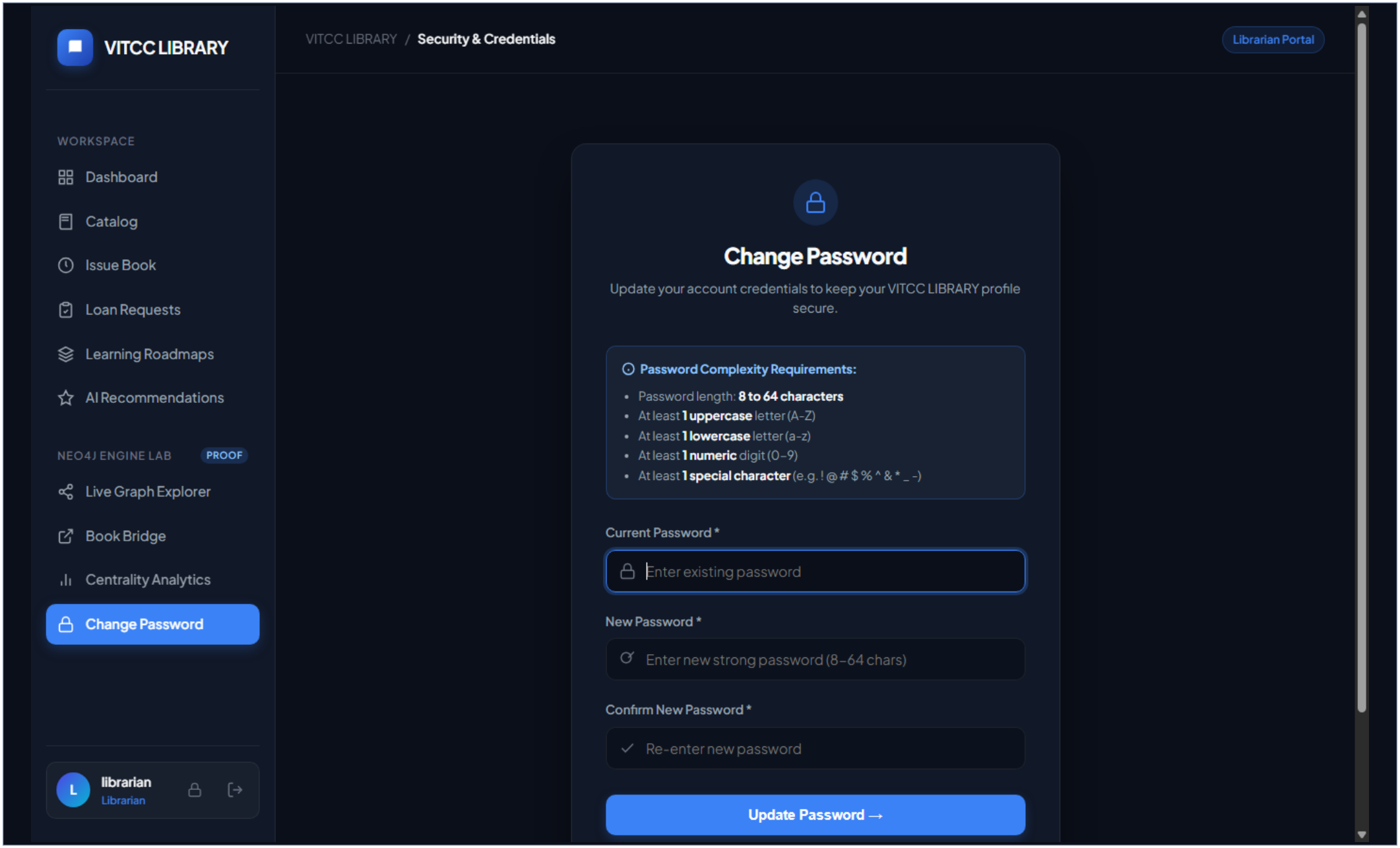
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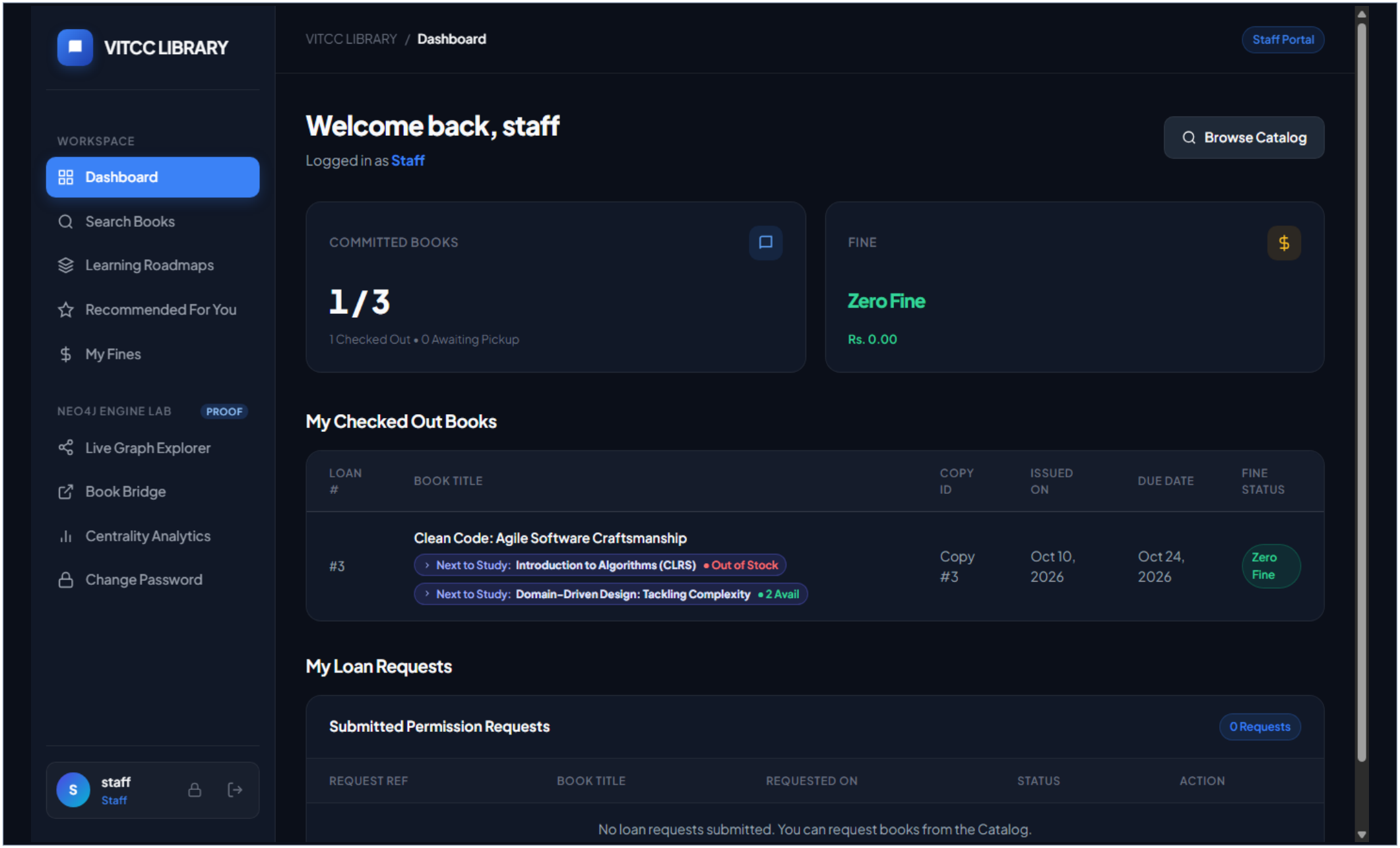
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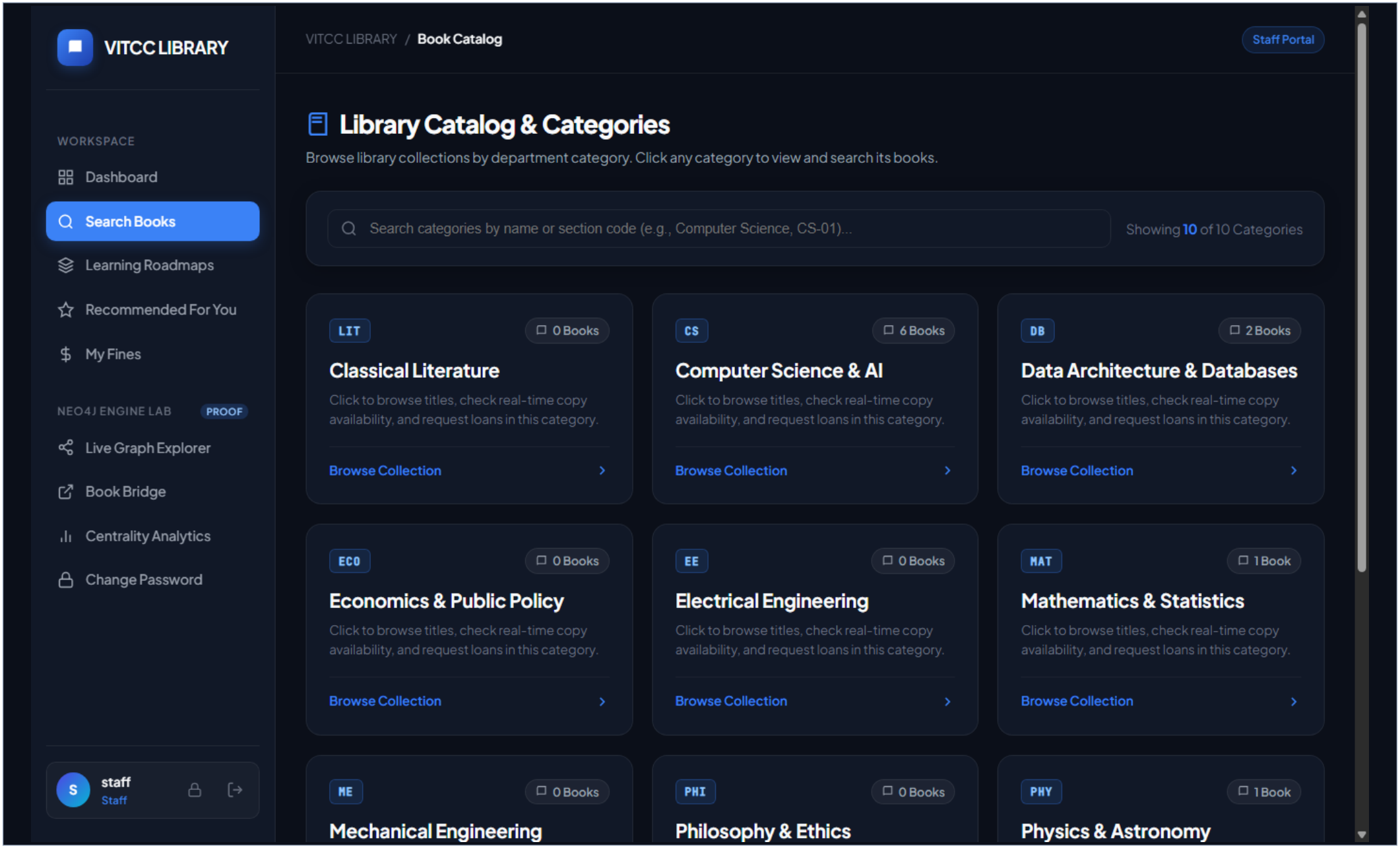
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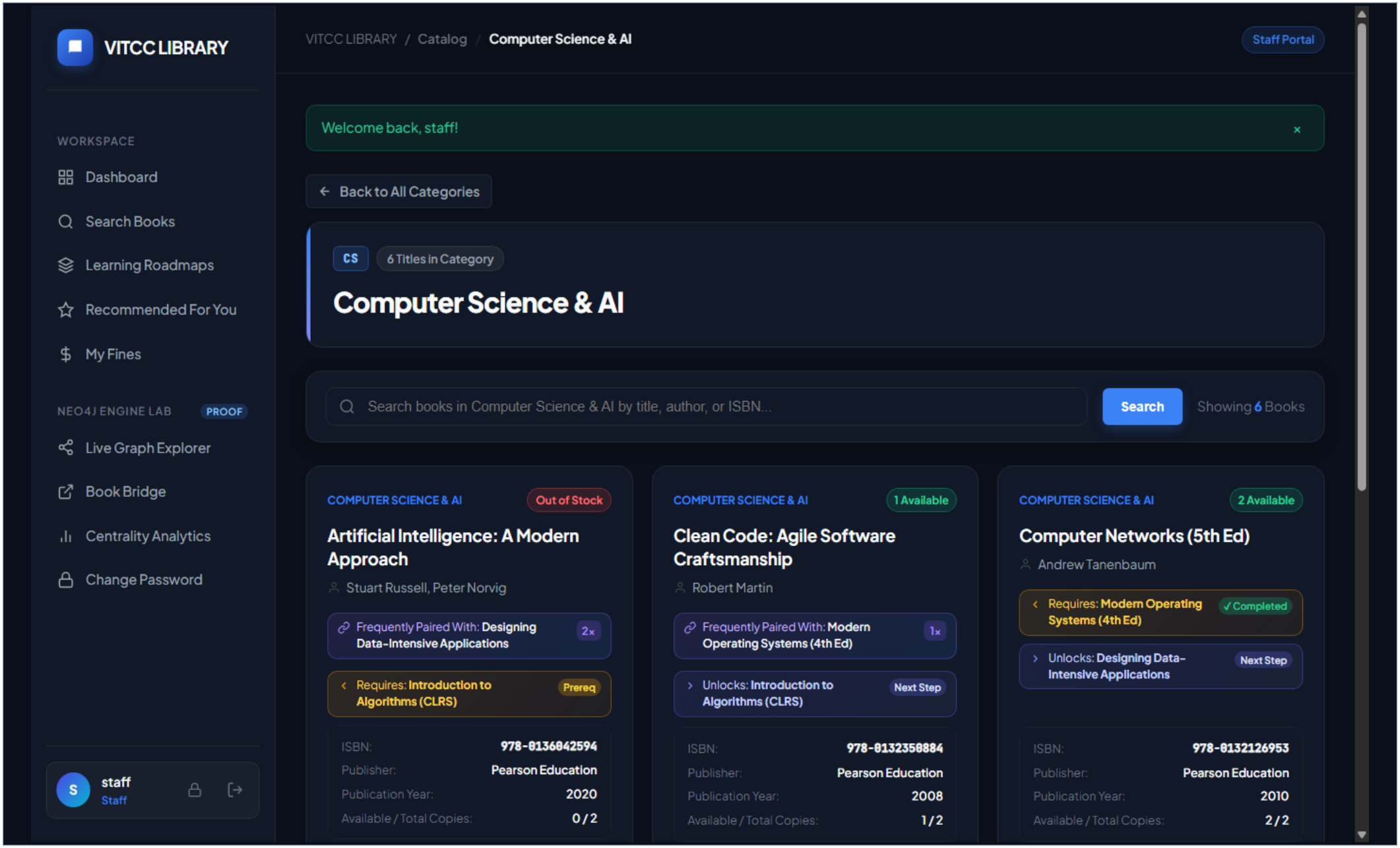


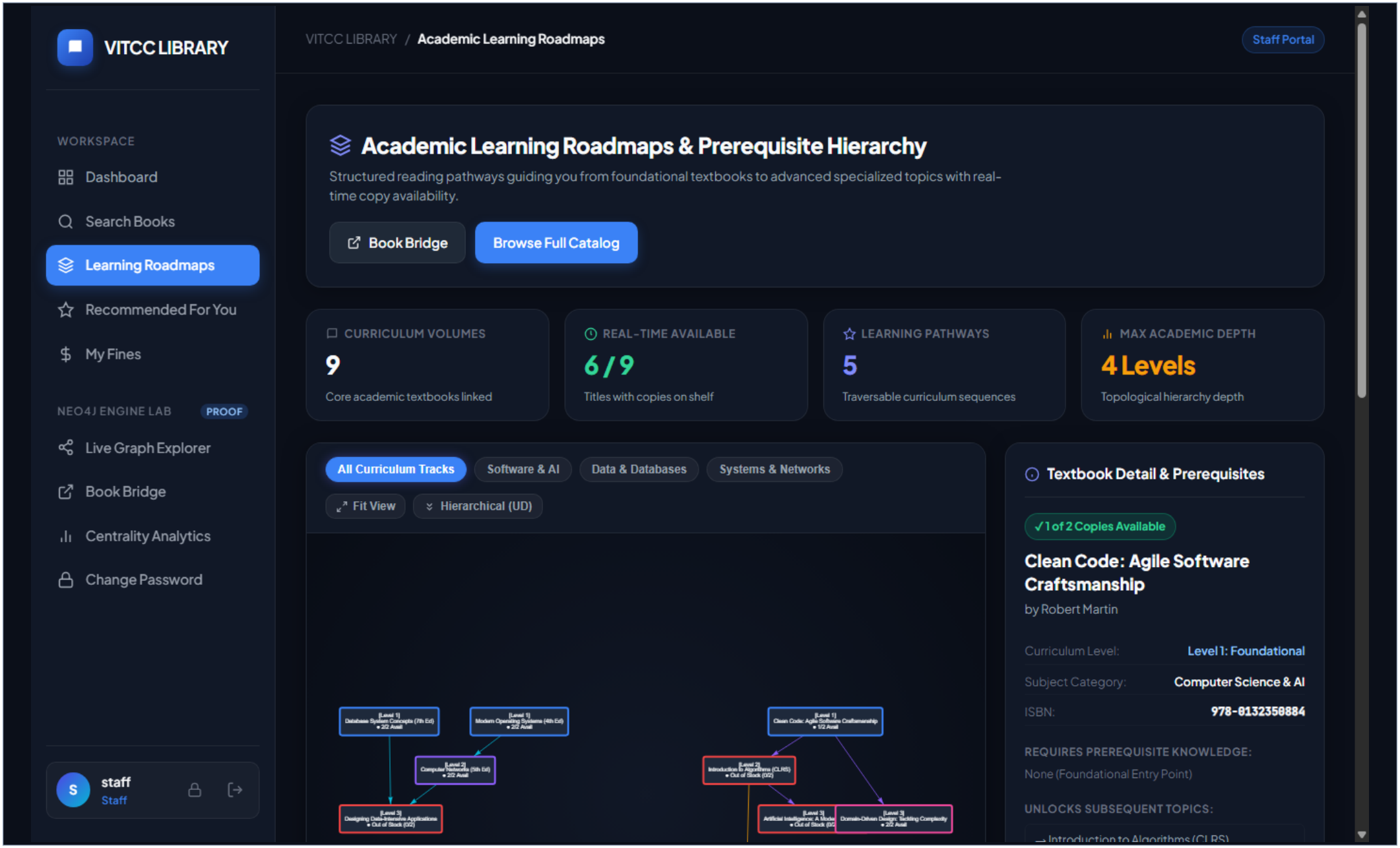

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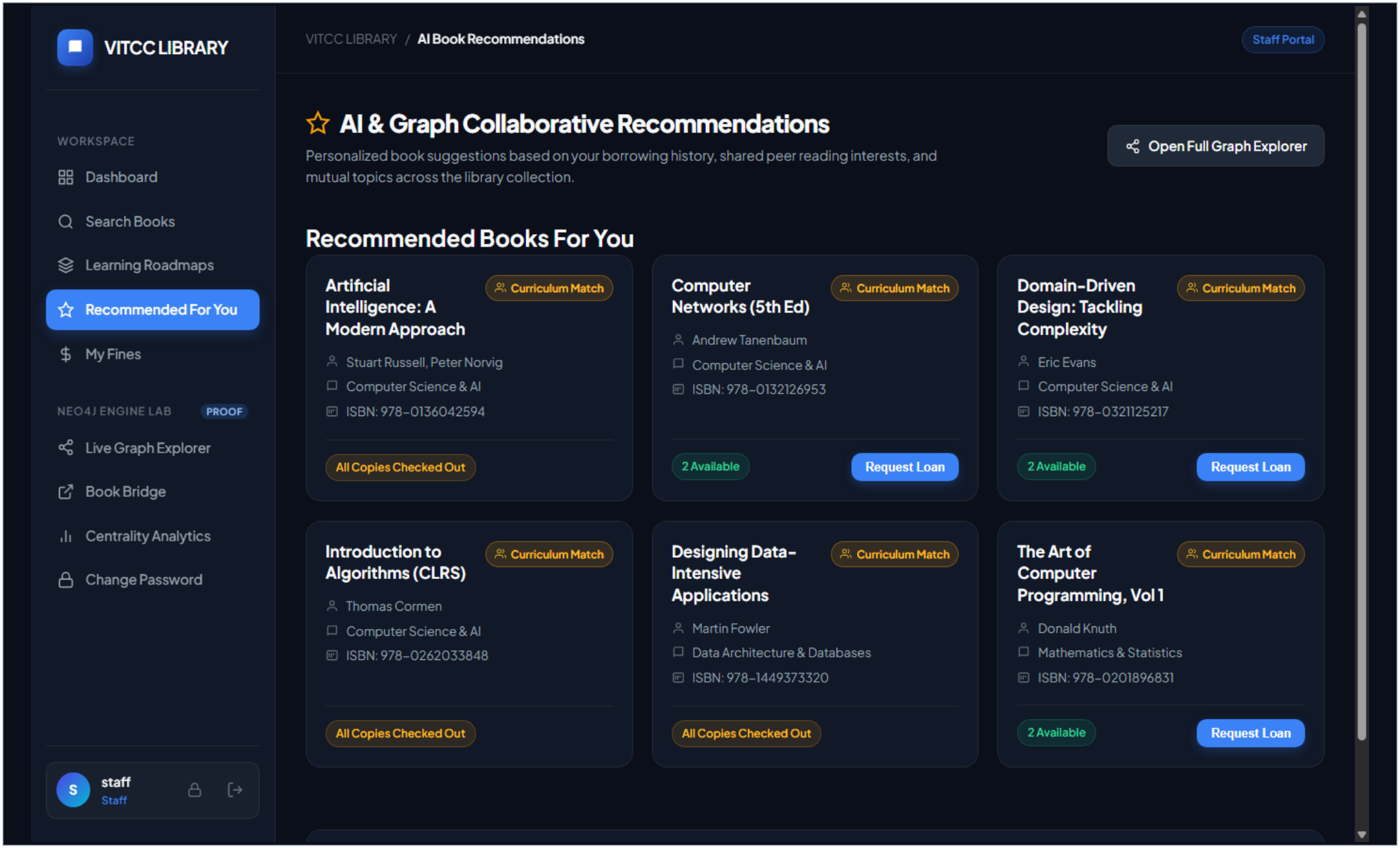

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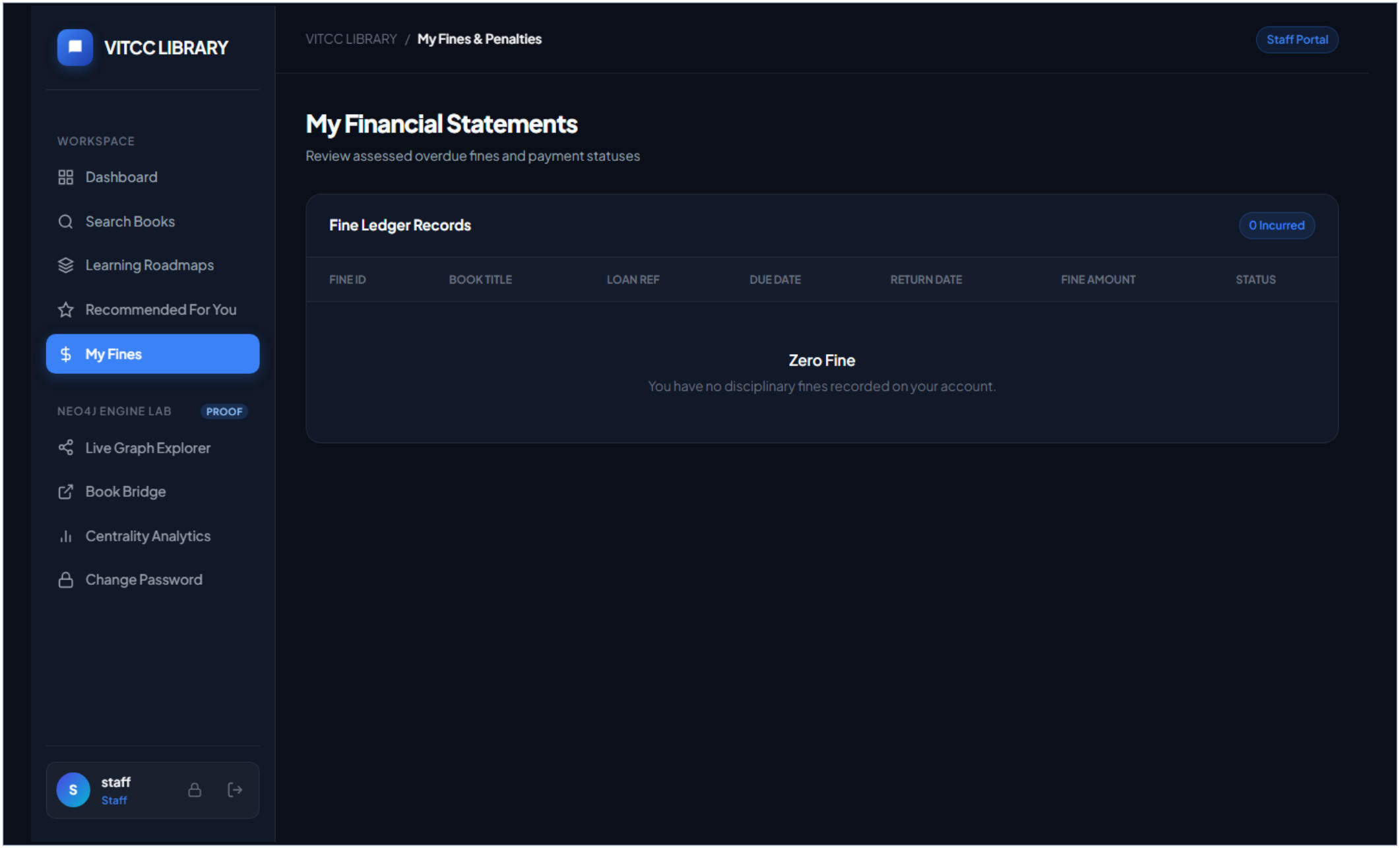

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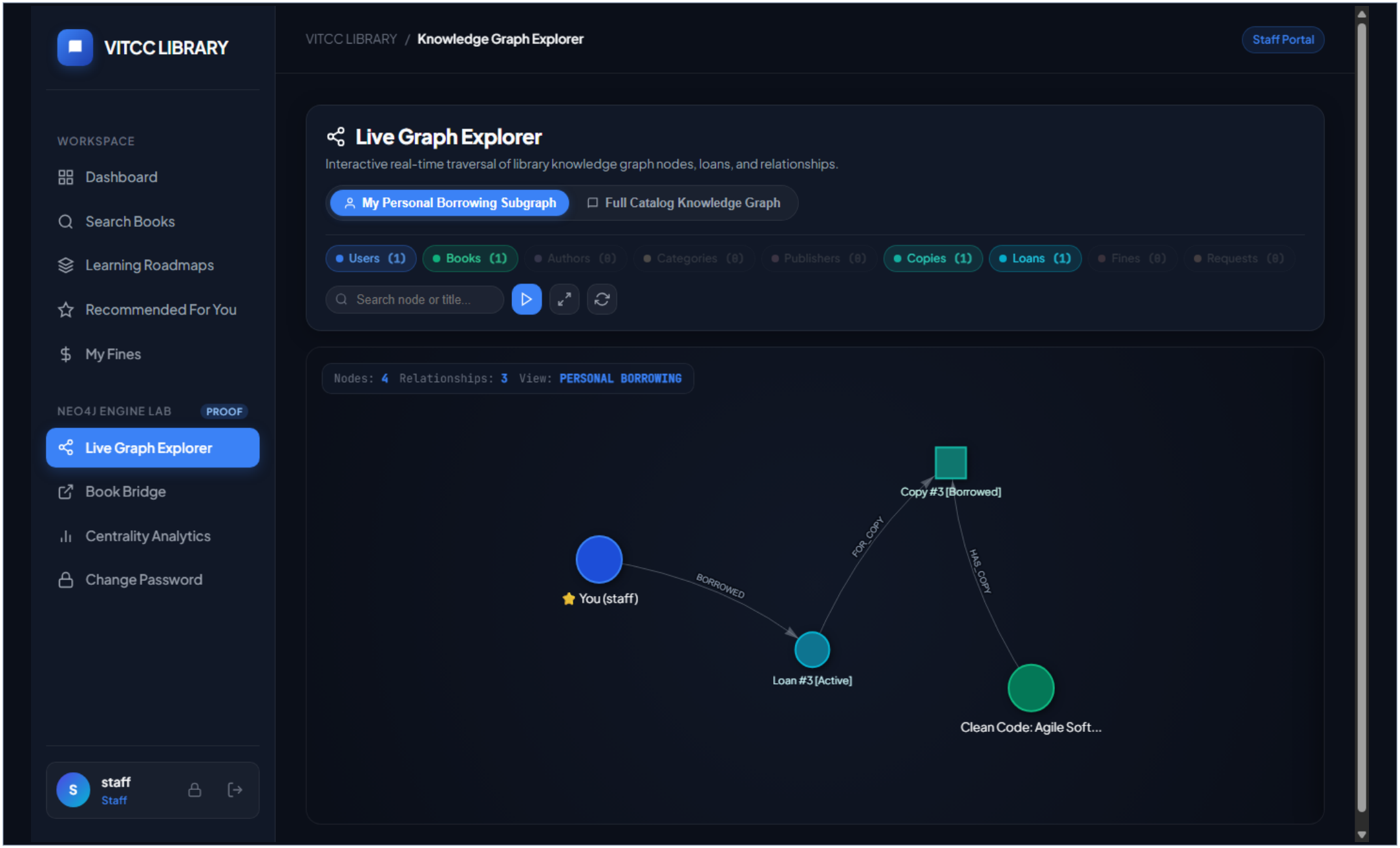
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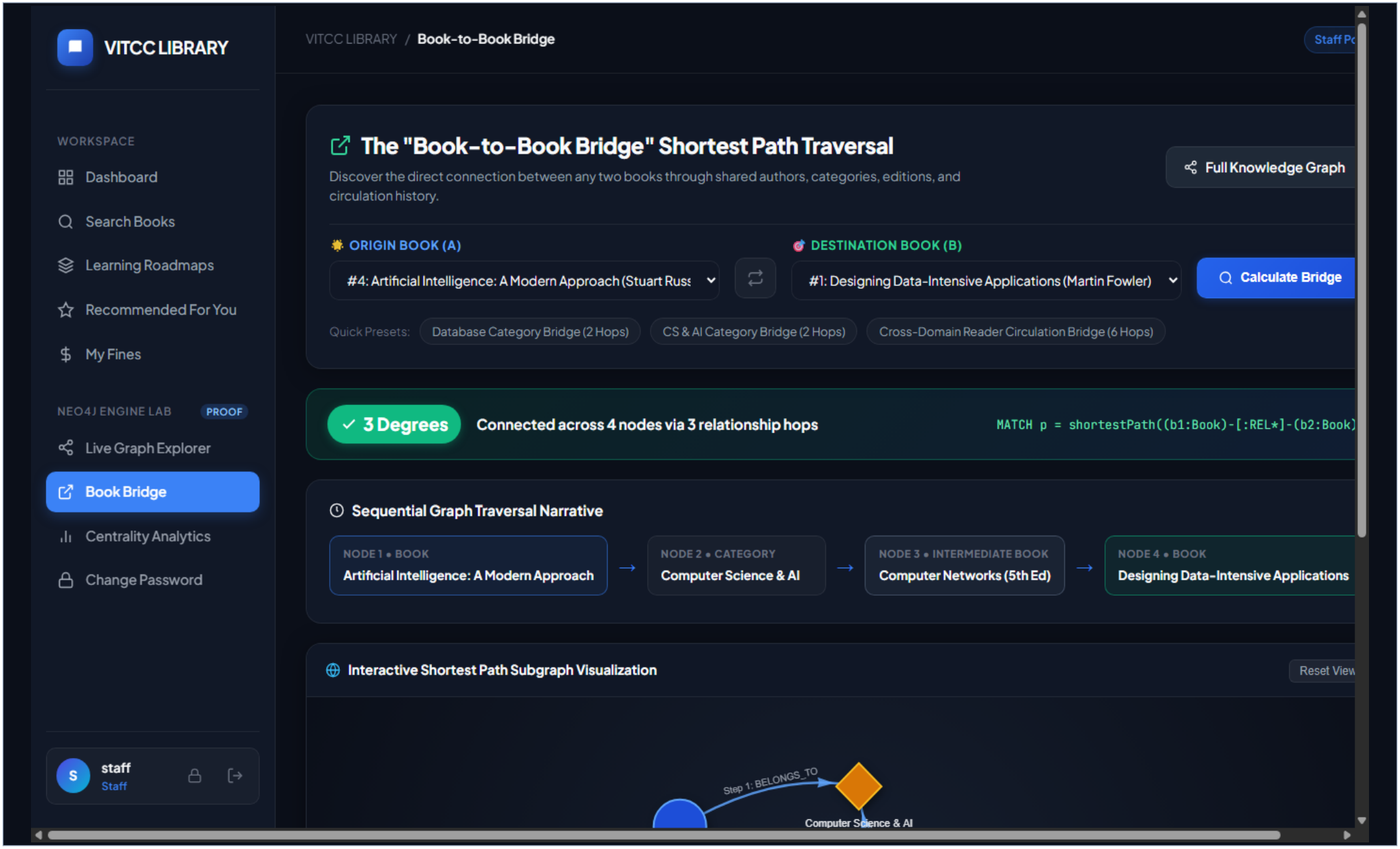
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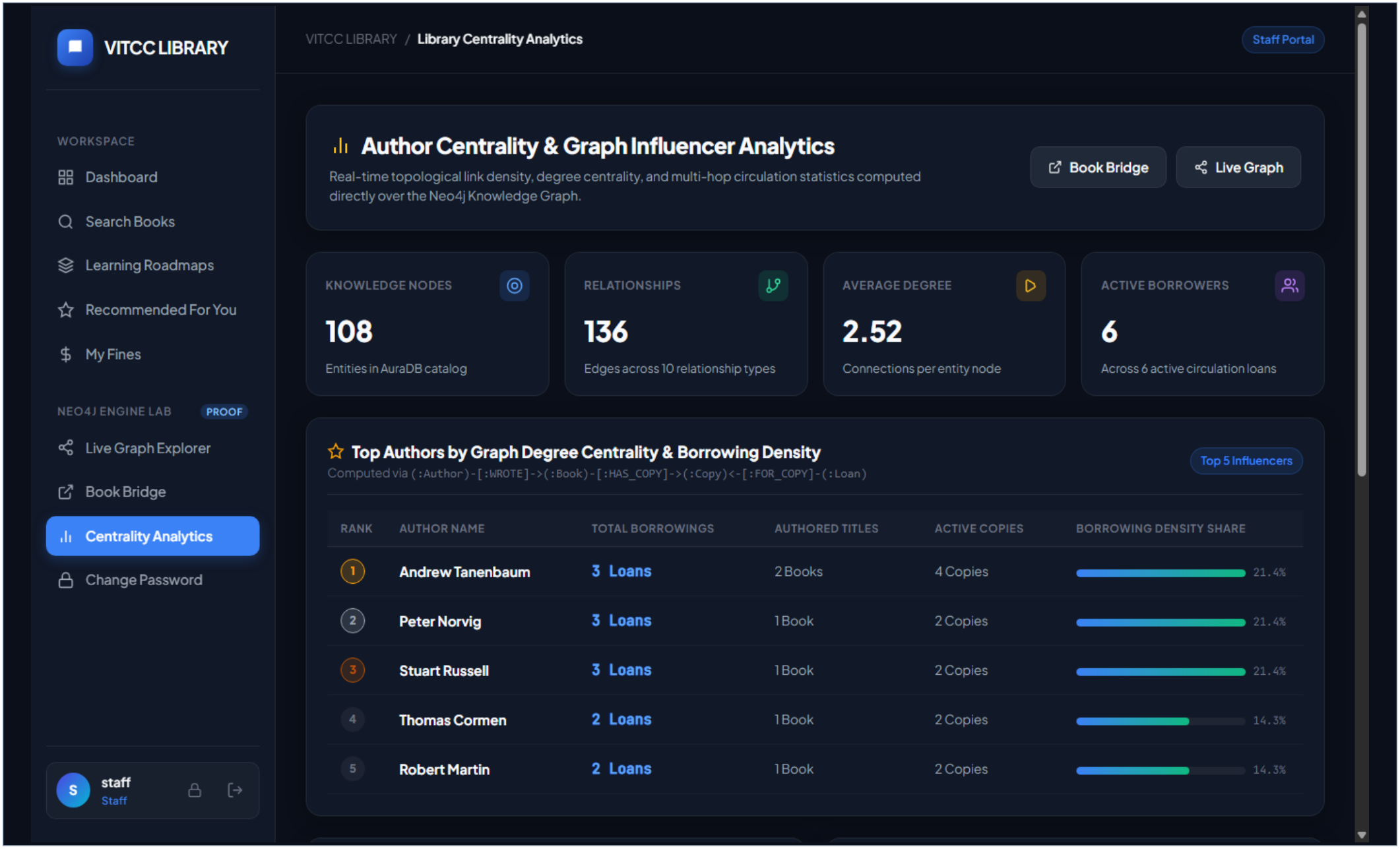
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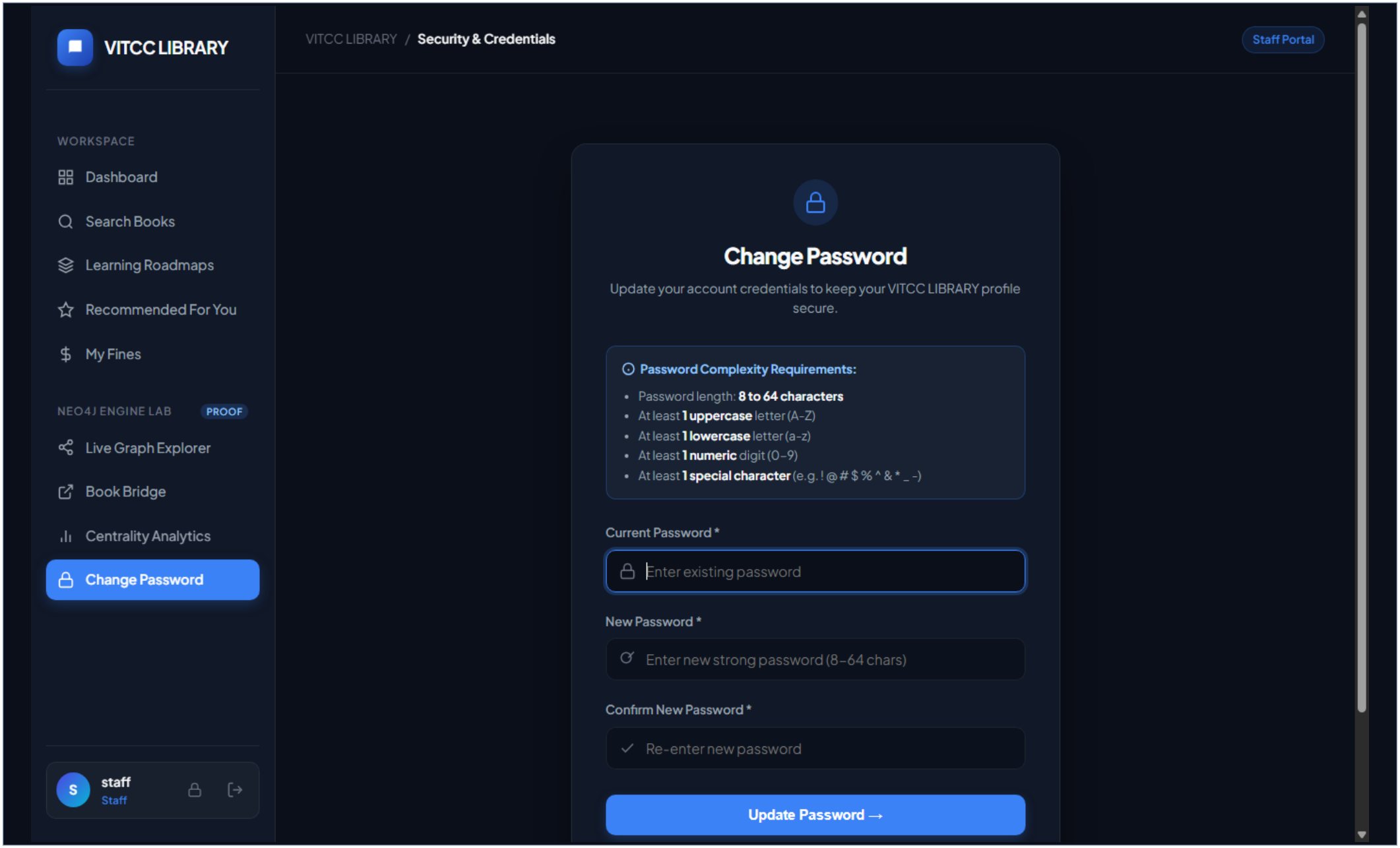
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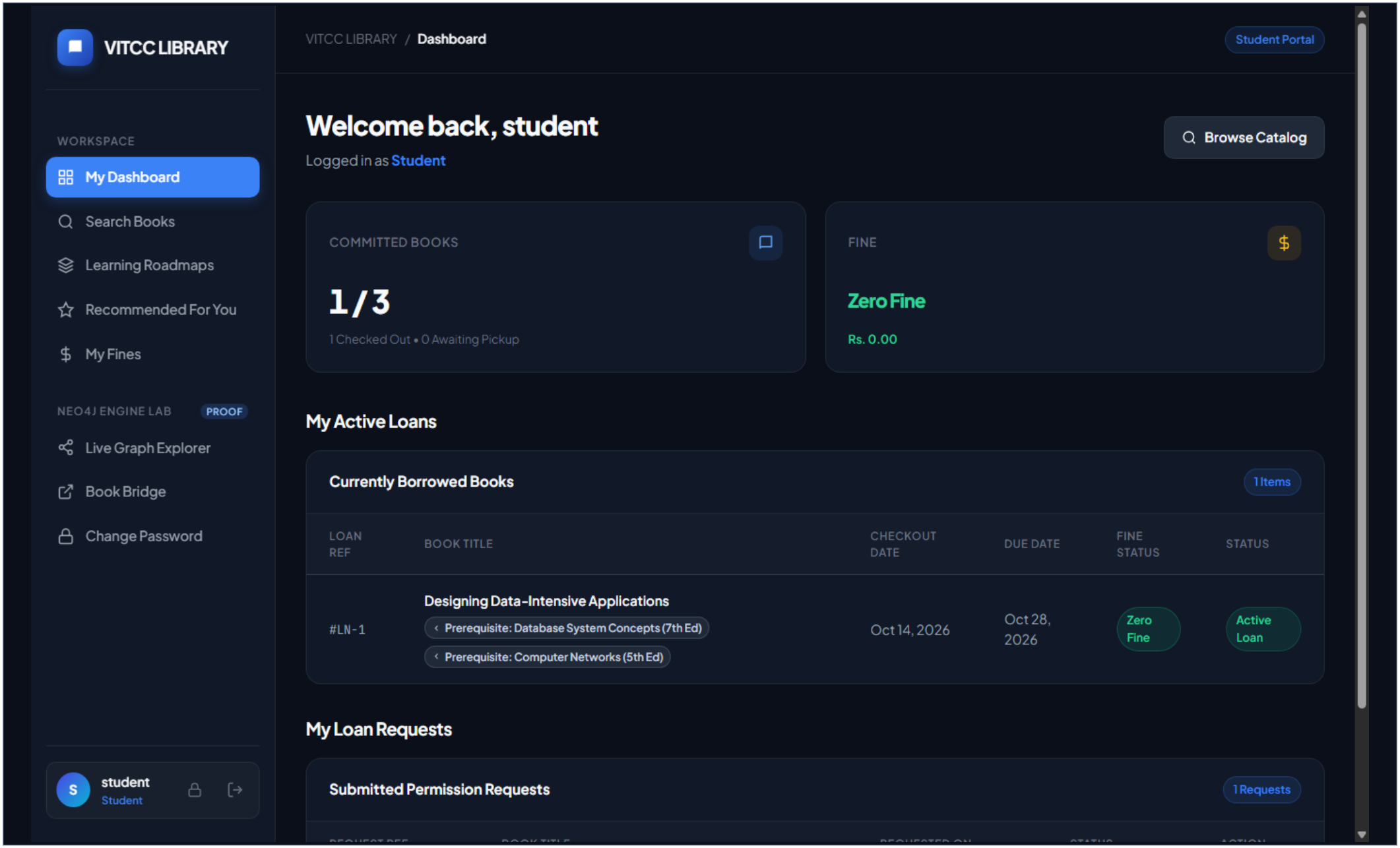

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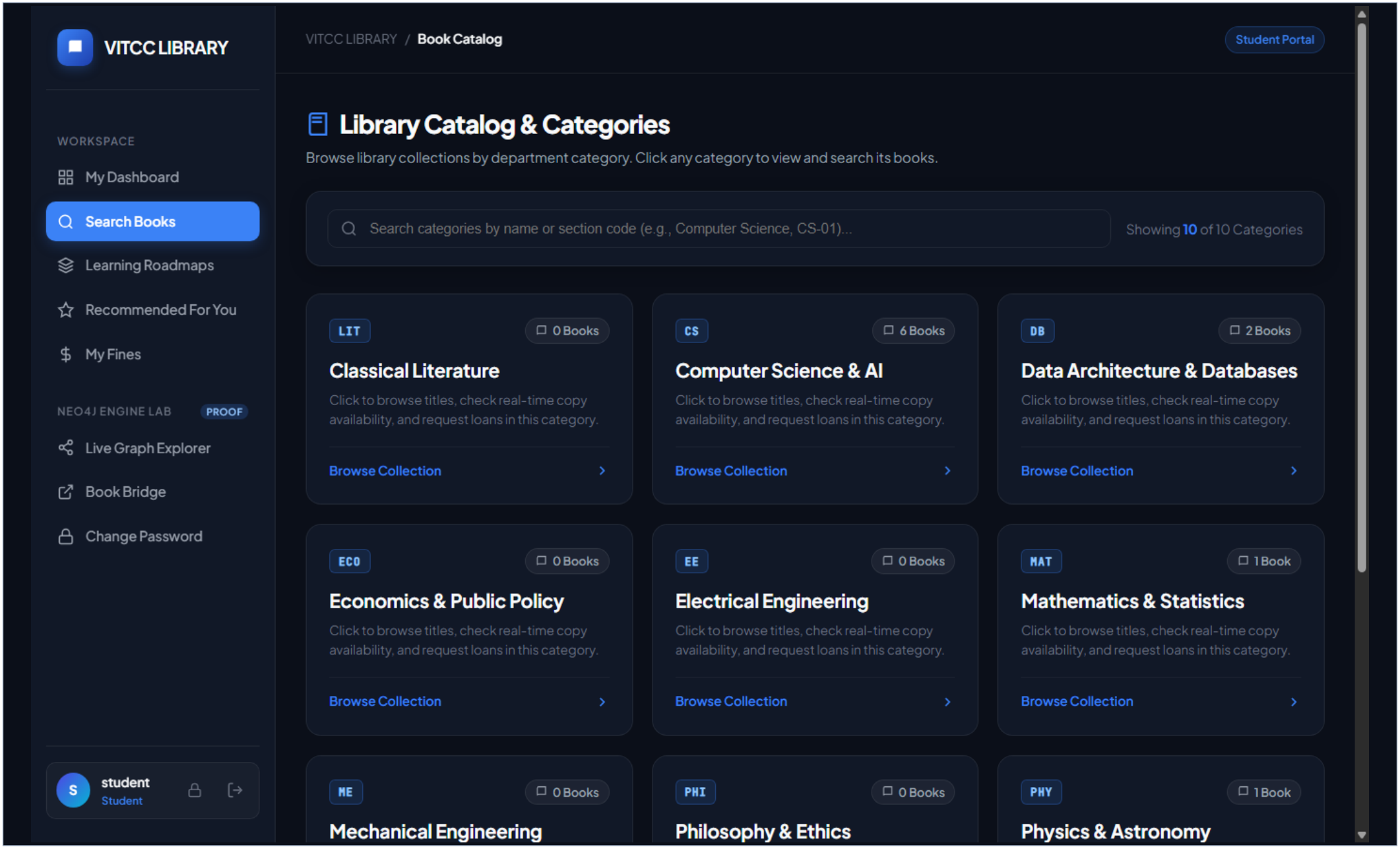
Student Academic Dashboard & Borrowings

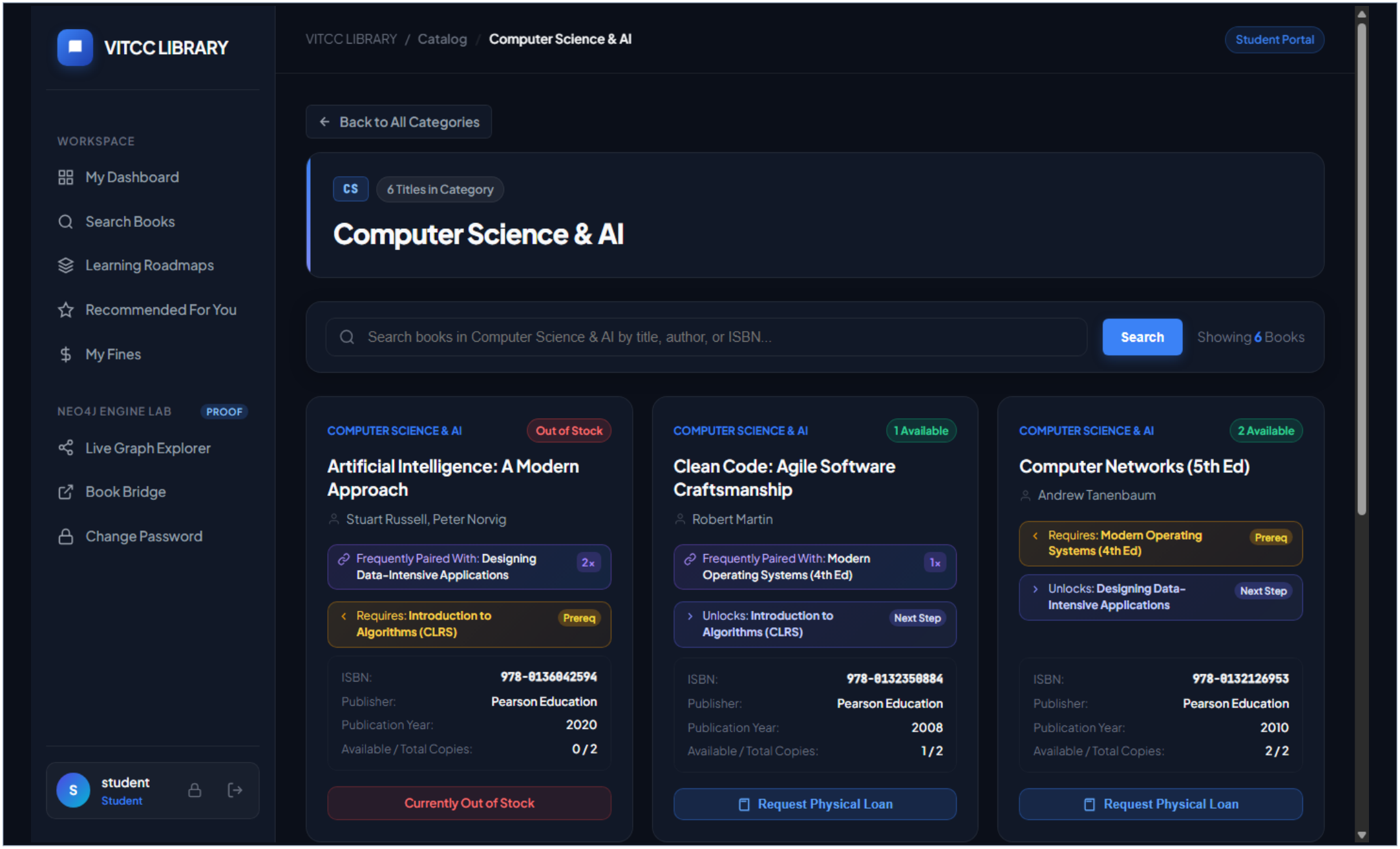
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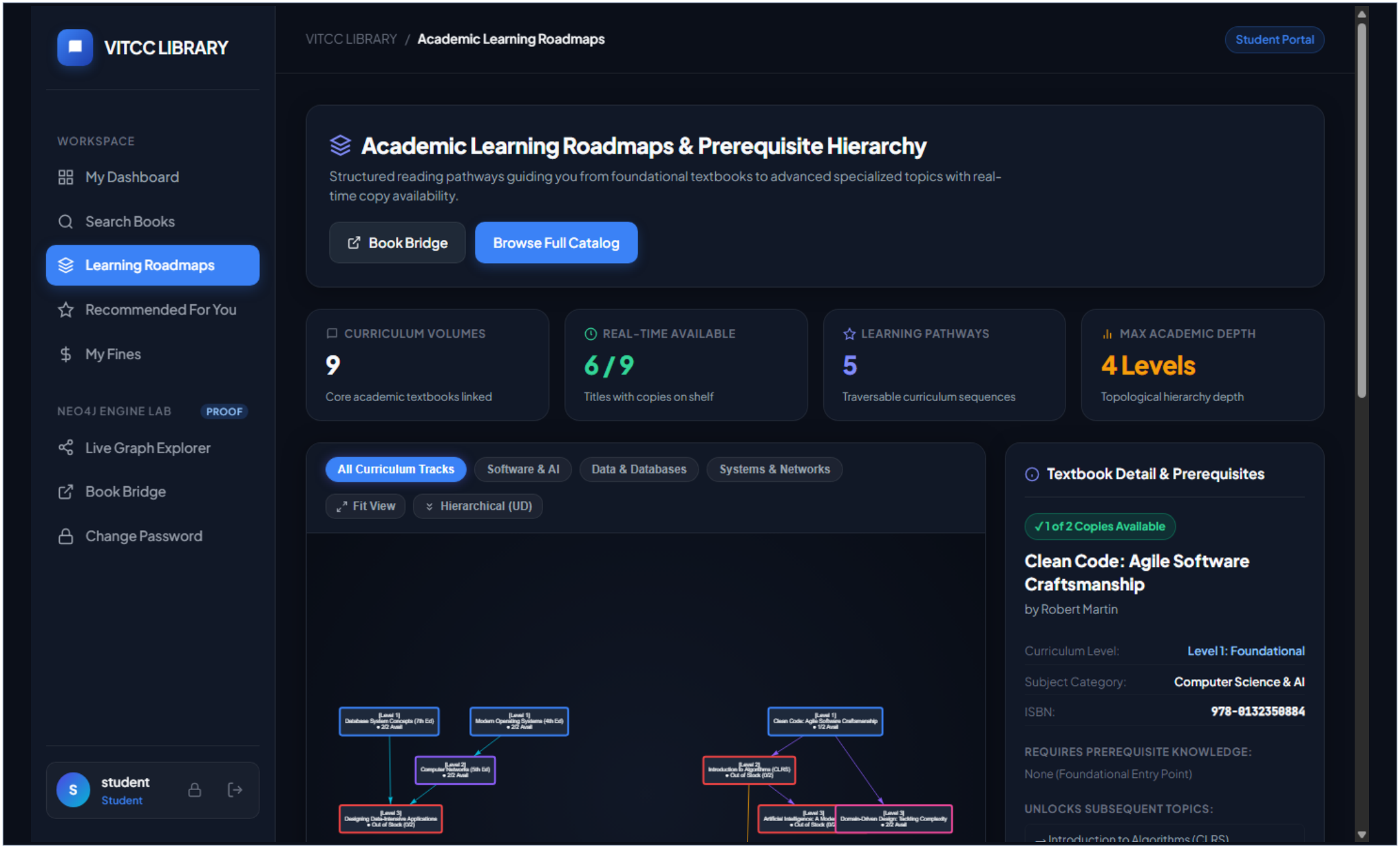
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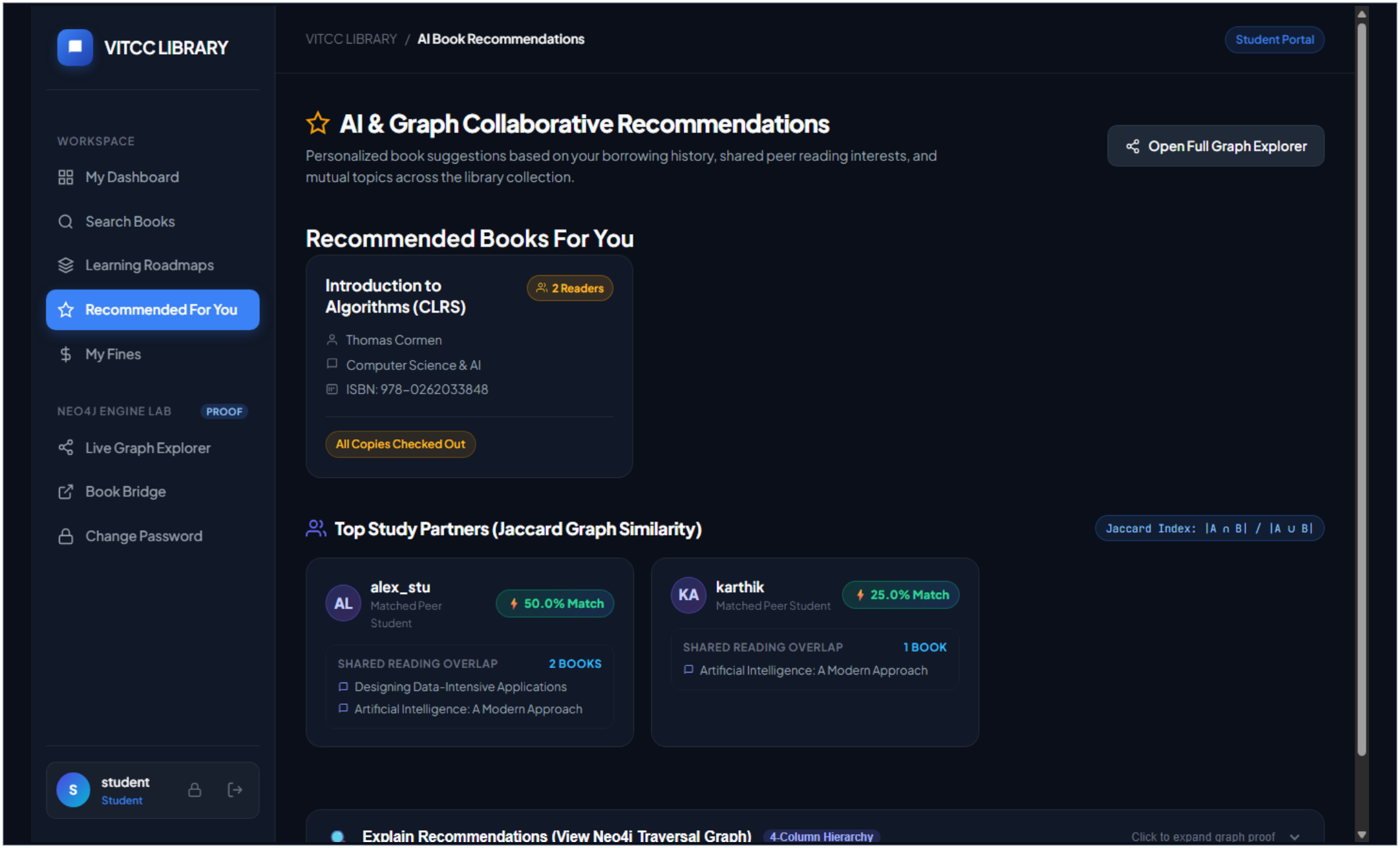
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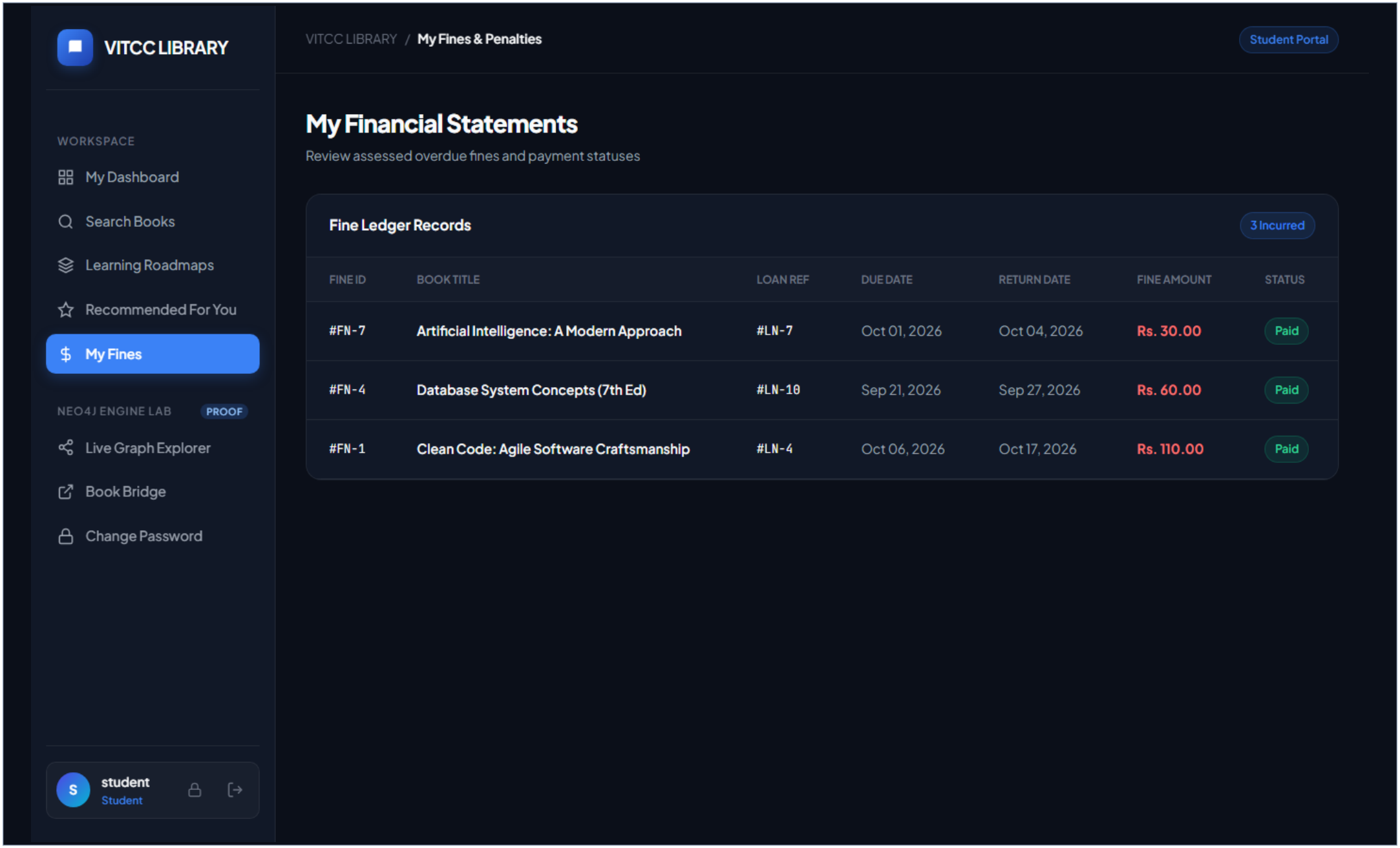

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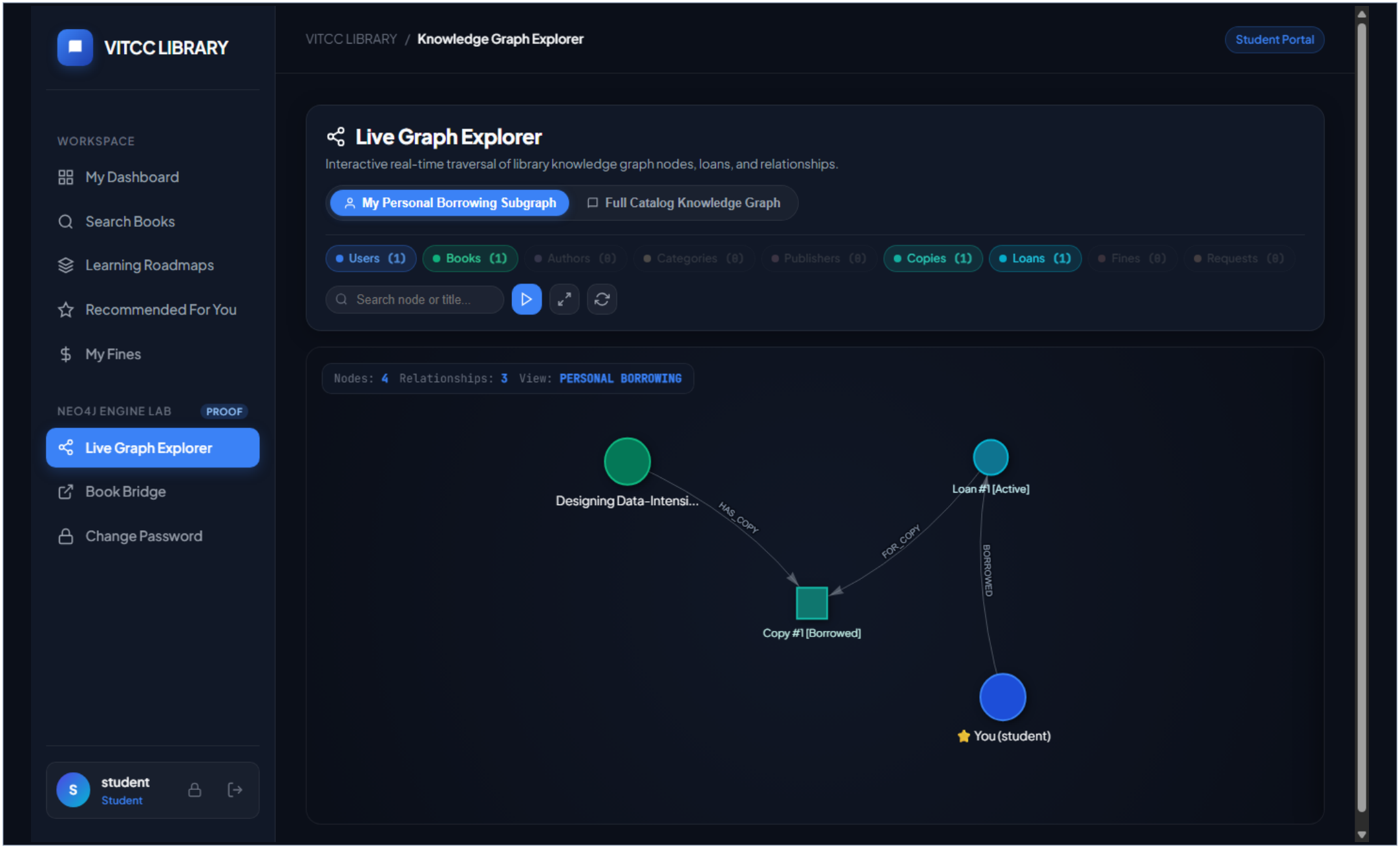
Personal Overdue Fines & Payment History

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Route: http://127.0.0.1:5000/my_fines | **ID:** 41_student_my_fines

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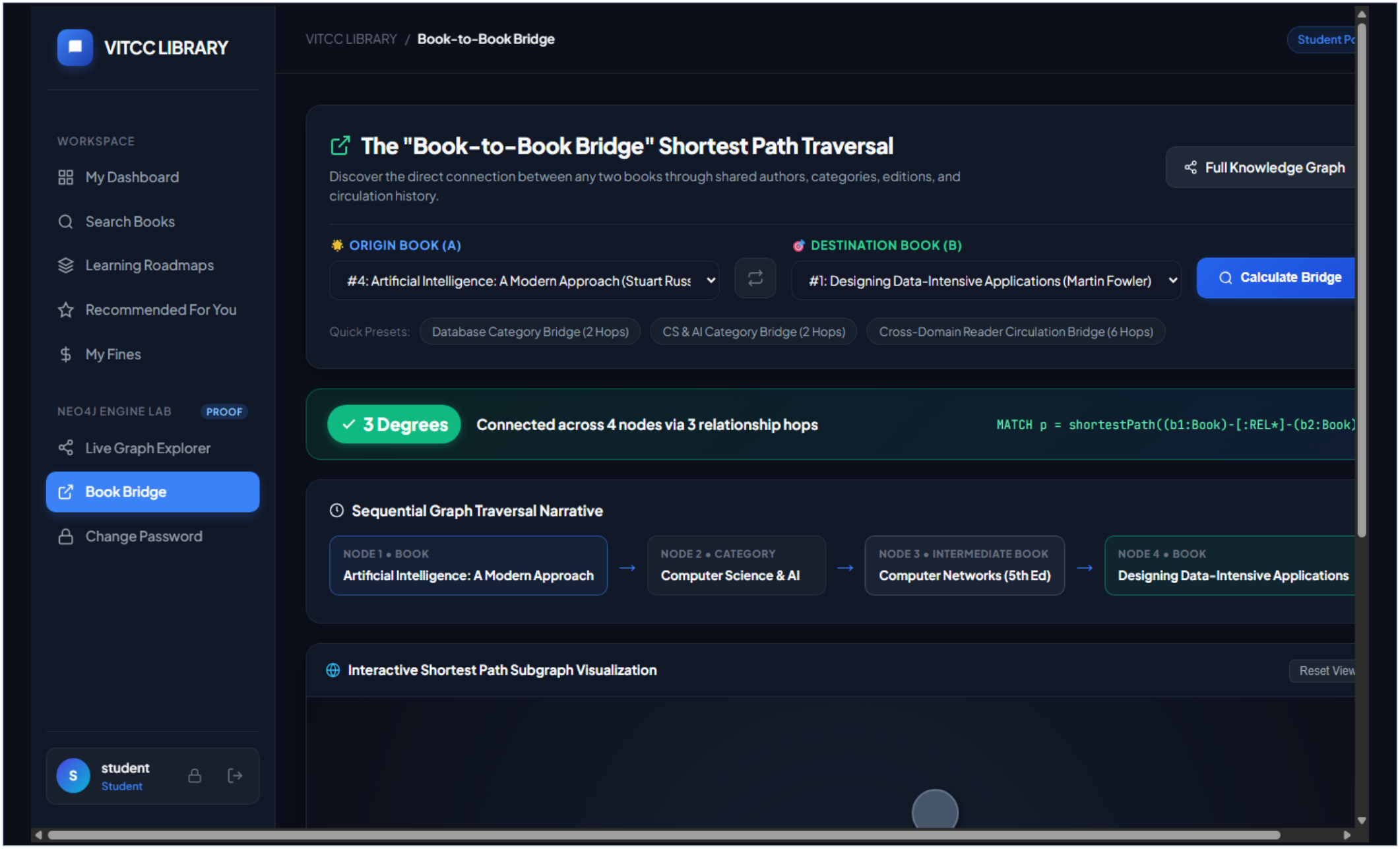

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